



Chapter 3 — Single Stage Amplifiers

Design of Analog CMOS
Integrated circuits - Behzad
Razavi

Presented by - Chinmay Joshi



CHAPTER OVERVIEW

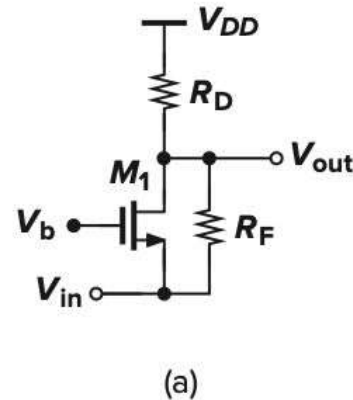
Concepts covered -

1. Common-Source Stage
 - a. CS Stage with Resistive Load
 - b. CS Stage with Diode Connected Load
 - c. CS Stage with Current Source Load
 - d. CS Stage with Triode Load
 - e. CS Stage with Source Degeneration
2. Source Follower (Common-Drain Stage)
3. Common-Gate Stage
4. Cascode Stage
 - a. Shielding Property
 - b. Folded Cascode

Problem 3.15-a

3.15. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.78 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

Diagram -



Handwritten Solution

3.15y

$V_{GS} = V_G - V_{in}$
 $V_{DS} = V_{out} - V_{in}$

For MOSFET to turn ON
 $V_{GS} > V_{th}$
 $V_{in} < V_G - V_{th}$

When $V_{in} = 0$, assuming V_G sets MOSFET in saturation
 $I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_G - V_{th})^2 \rightarrow \text{finite}$
 $V_{out} \rightarrow \text{finite}$

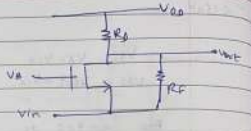
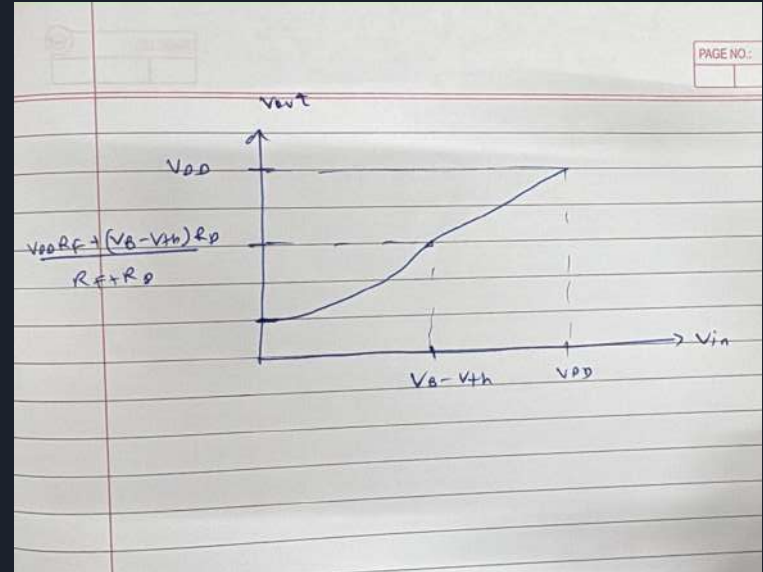
At boundary: $V_{in} = V_G - V_{th}$, Assuming MOSFET EPP

$V_{DD} - V_{out} = \frac{V_{out} - V_{in}}{R_D}$
 $V_{in} = V_G - V_{th}$

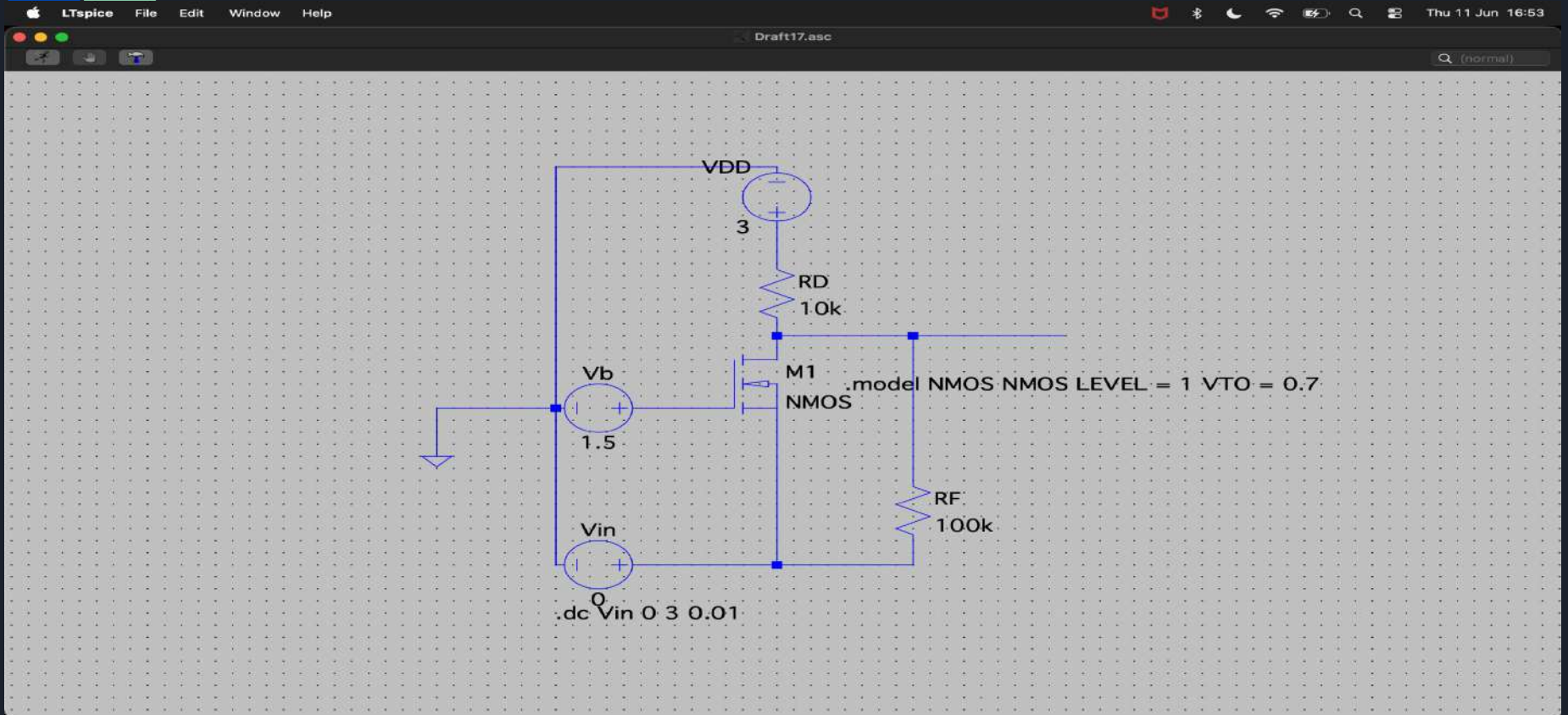
$V_{out} = \frac{V_{DD} R_D + R_D (V_G - V_{th})}{R_D + R_D}$

After $V_{in} = V_G - V_{th}$, MOSFET turns off and V_{out} is only affected by R_D and R_D .

When $V_{in} = V_{DD}$, current becomes 0 and $V_{out} = V_{DD}$

LTSPICE Schematic



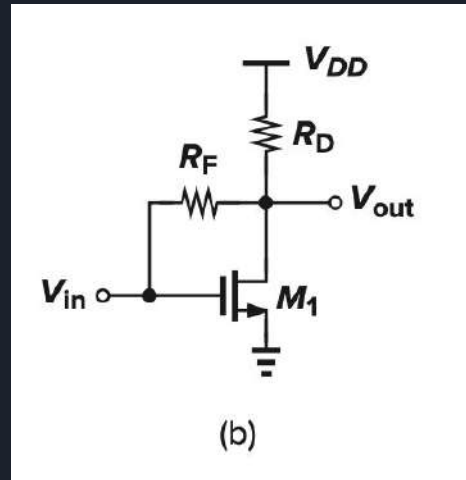
Simulation Results for Vout



Problem 3.15-b

3.15. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.78 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

Diagram -



Handwritten Solution

3-15-16

When $V_{in} < V_{th}$

MOSFET is OFF

$$V_{out} = \frac{V_{DD}R_F + V_{in}R_D}{R_F + R_D}$$

Once $V_{in} > V_{th}$, MOSFET turns ON in saturation

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{th})^2$$

$$I_{RF} + I_{DD} = I_D$$

Initially I_D is low (and) compared to I_{RF} (and)

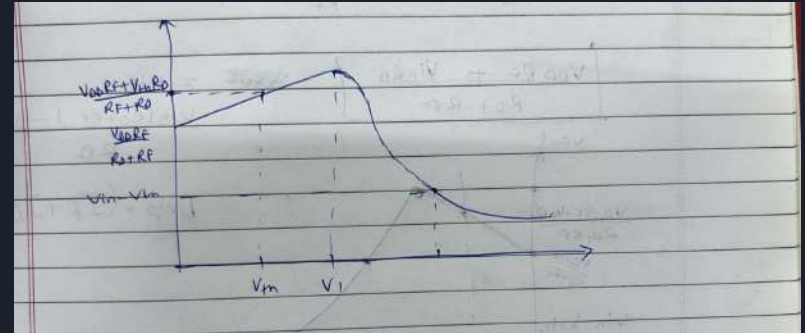
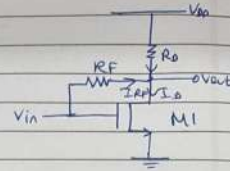
as $V_{in} \uparrow$, $I_{RF} \downarrow$, therefore $V_{out} \uparrow$

After $V_{in} = V_1$, $\frac{dV_{out}}{dV_{in}} = 0$

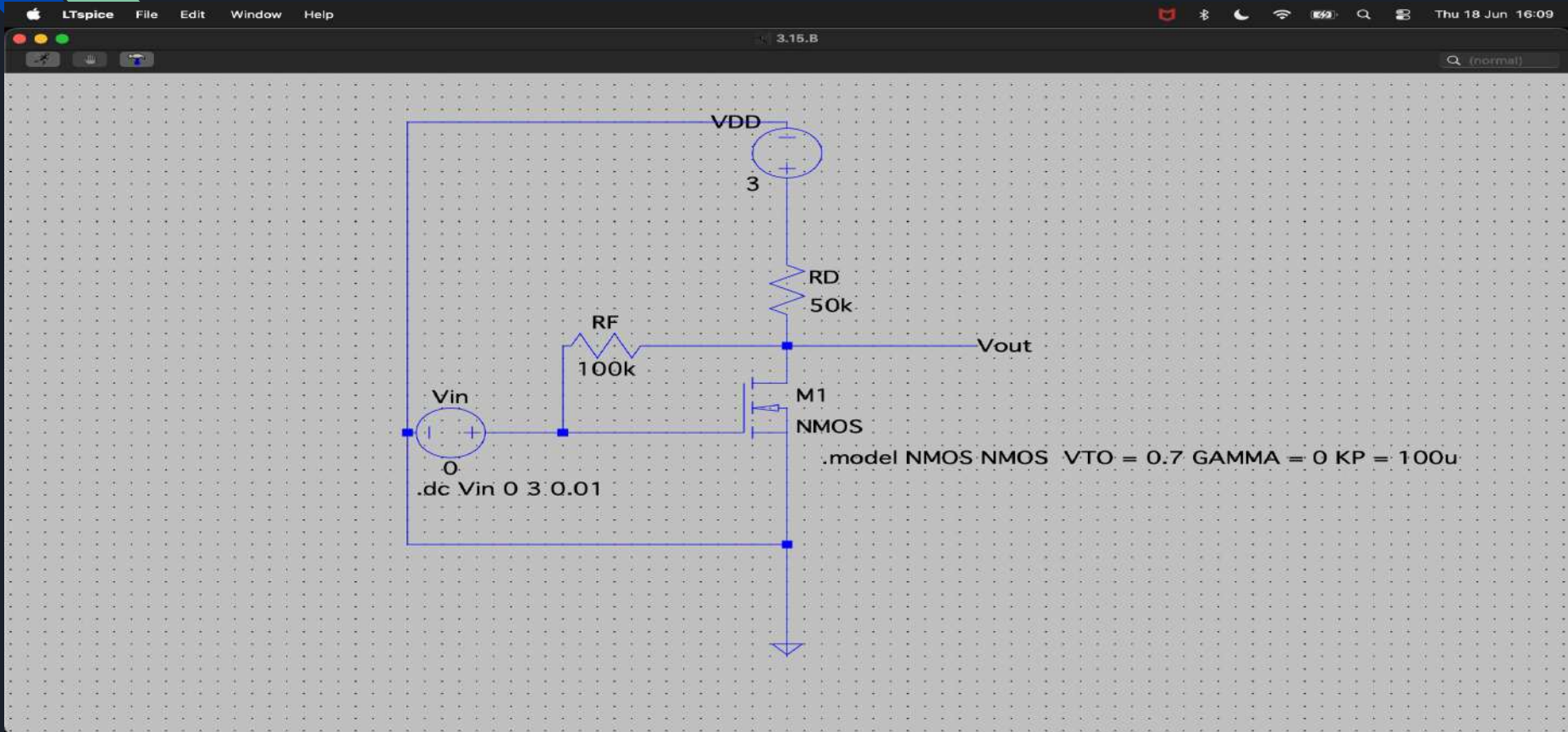
I_D becomes large enough and as $V_{in} \uparrow$, $I_{DD} \uparrow$, V_{out} decreases

When $V_{out} = V_{in} - V_{th}$ saturation-triode transition occurs.

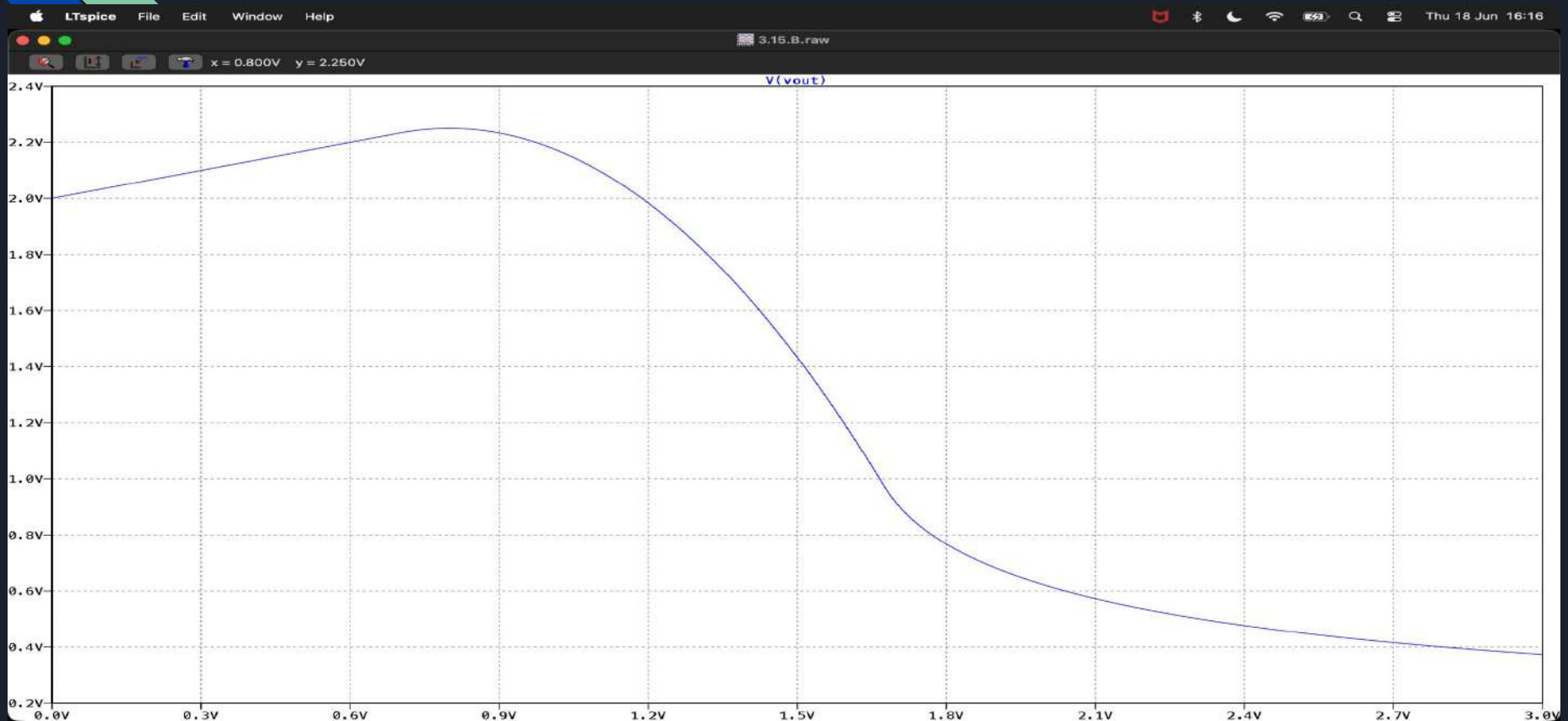
$\therefore$ MOSFET enters triode region for $V_{out} < V_{in} - V_{th}$



LTSPICE Schematic



Simulation Results for Vout





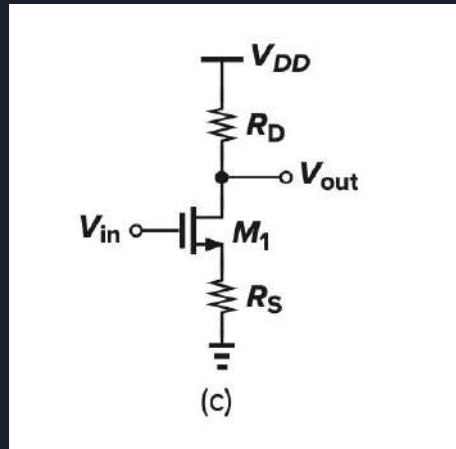
Observations

- Ideally, we expect V_{out} to start decreasing immediately after $V_{in} > V_{th}$, as the MOSFET starts conducting and pulling current through the load.
- However, in the practical case, for $0.7V < V_{in} < 0.8V$, the MOSFET drain current I_D is still very small compared to the reference current I_{RF} . Therefore, the voltage drop across the load is small, causing V_{out} to continue increasing until approximately $V_{in} = 0.8V$.
- Beyond $V_{in} = 0.8V$, I_D increases significantly, resulting in a larger voltage drop across the load resistor and causing V_{out} to decrease.
- The MOSFET enters the triode region when: $V_{out} < V_{in} - V_{th}$

Problem 3.15-c

3.15. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.78 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

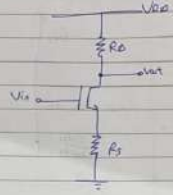
Diagram -



Handwritten Solution

2.16 VCB

This is a CS amplifier with Source Degeneration



① When $V_{in} < V_{th}$

MOSFET is OFF, no current in C_K

$V_{out} = V_{DD}$

② When $V_{in} > V_{th}$, V_{out} is high initially

$V_{DS} = V_{out} - V_S$
 $V_{GS} = V_{in} - V_S$

MOSFET will be in saturation

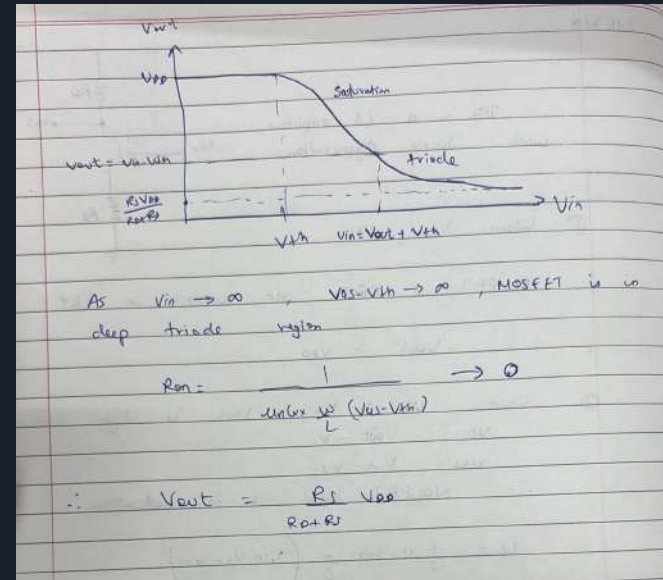
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_S - V_{th})^2$$

MOSFET is in saturation & u

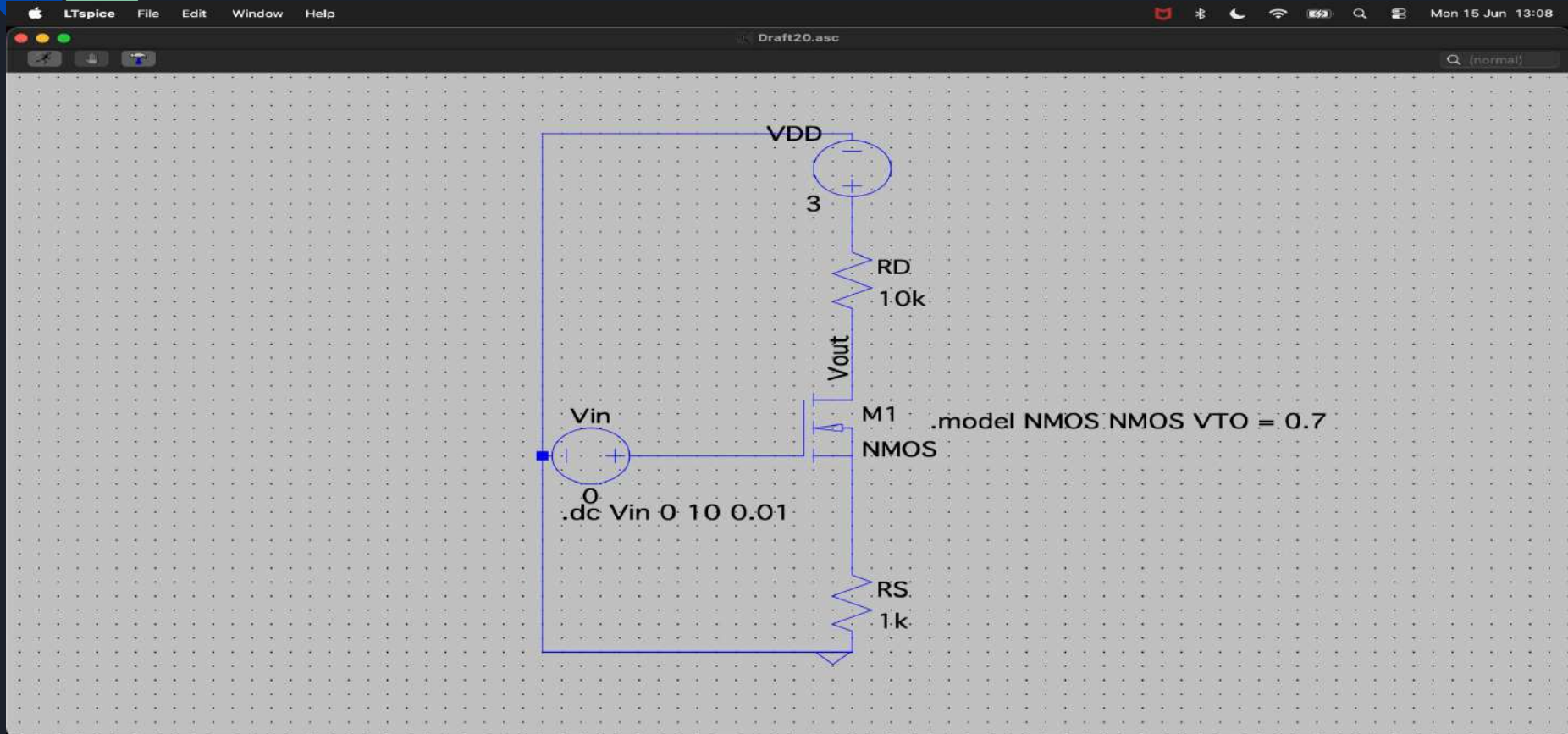
$V_{GS} = V_{GS} - V_{th}$
 $V_{out} = V_{in} - V_{th}$

As V_{in} increases, I_D increases, drop across R_D increases causing V_{out} to decrease

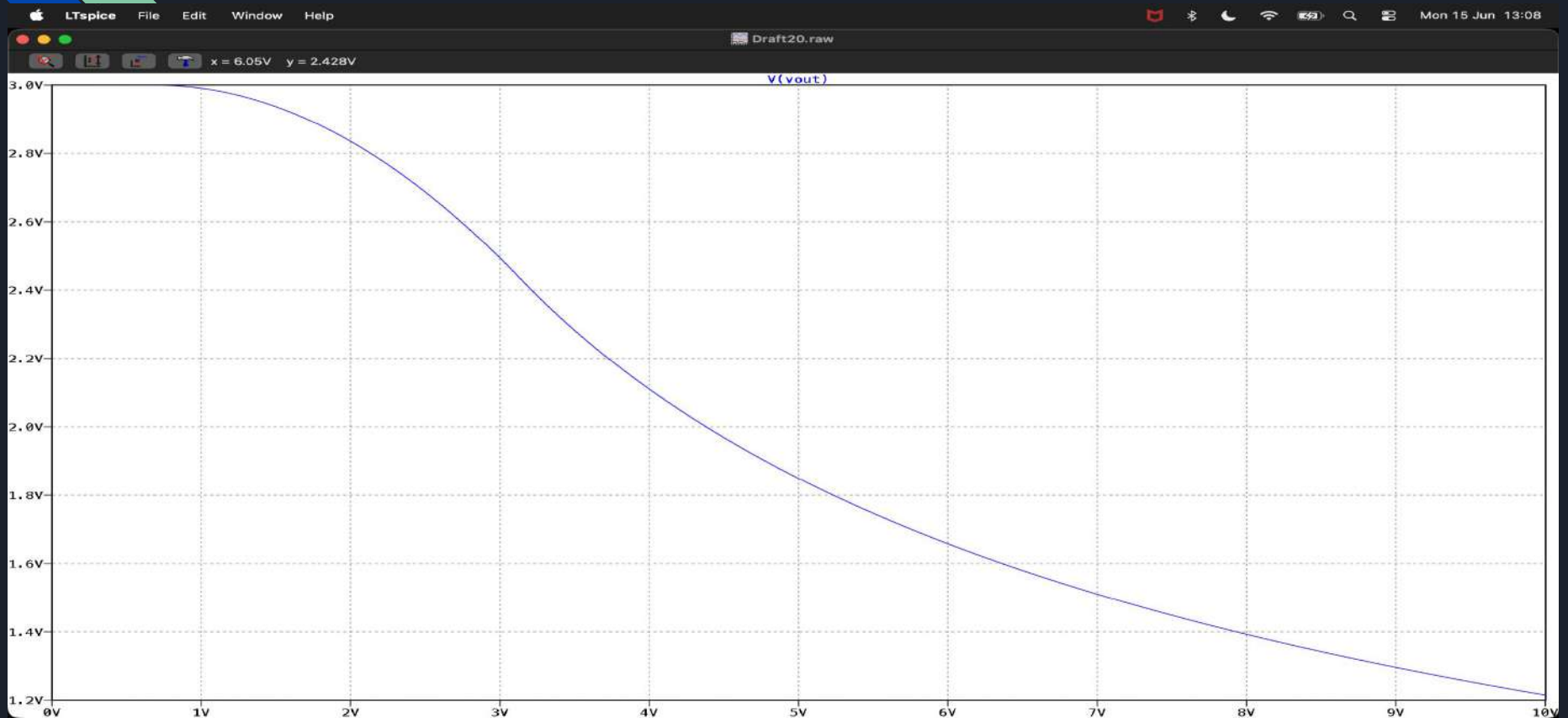
After $V_{in} = V_{out} + V_{th}$, MOSFET enters triode region.



LTSPICE Schematic



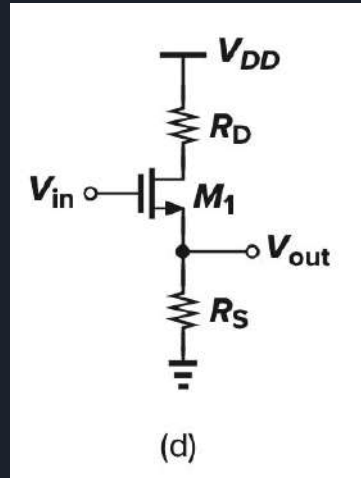
Simulation Results for Vout



Problem 3.15-d

3.15. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.78 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

Diagram -

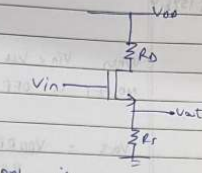


Handwritten Solution

Ex 4.4

When $V_{in} < V_{th}$, MOSFET is OFF

$$V_{out} = 0$$



Once $V_{in} > V_{th}$, MOSFET turns ON in saturation

$$V_{GS} = V_{in} - V_{out}, \quad V_{DS} = V_{DD} - I_D R_D - V_{out}$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

$$V_{out} = I_D R_S$$

As $V_{in} \uparrow$, $I_D \uparrow$, $V_{out} \uparrow$

When $V_{GS} - V_{th} > V_{DS}$

$$V_{in} - V_{out} - V_{th} > V_{DD} - I_D R_D - V_{out}$$

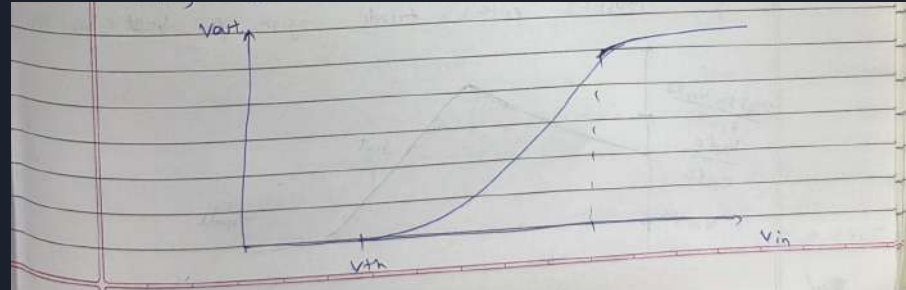
$$V_{in} > V_{DD} - I_D R_D + V_{th} \quad \text{MOSFET}$$

enters triode region

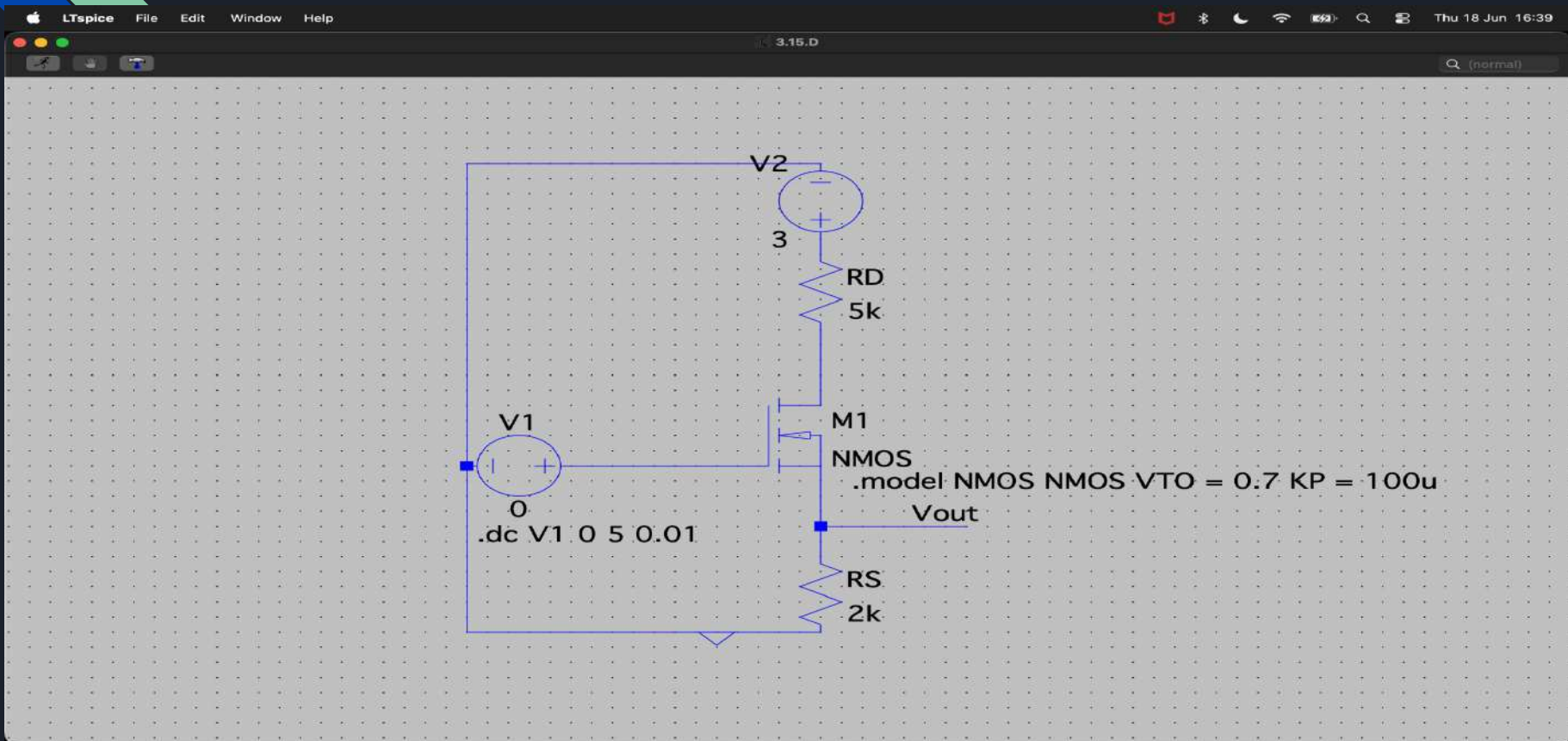
$$\Rightarrow \text{at boundary } V_{in} = V_{DD} - I_D R_D + V_{th} \quad \text{--- (1)}$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - I_D R_S - V_{th})^2 \quad \text{--- (2)}$$

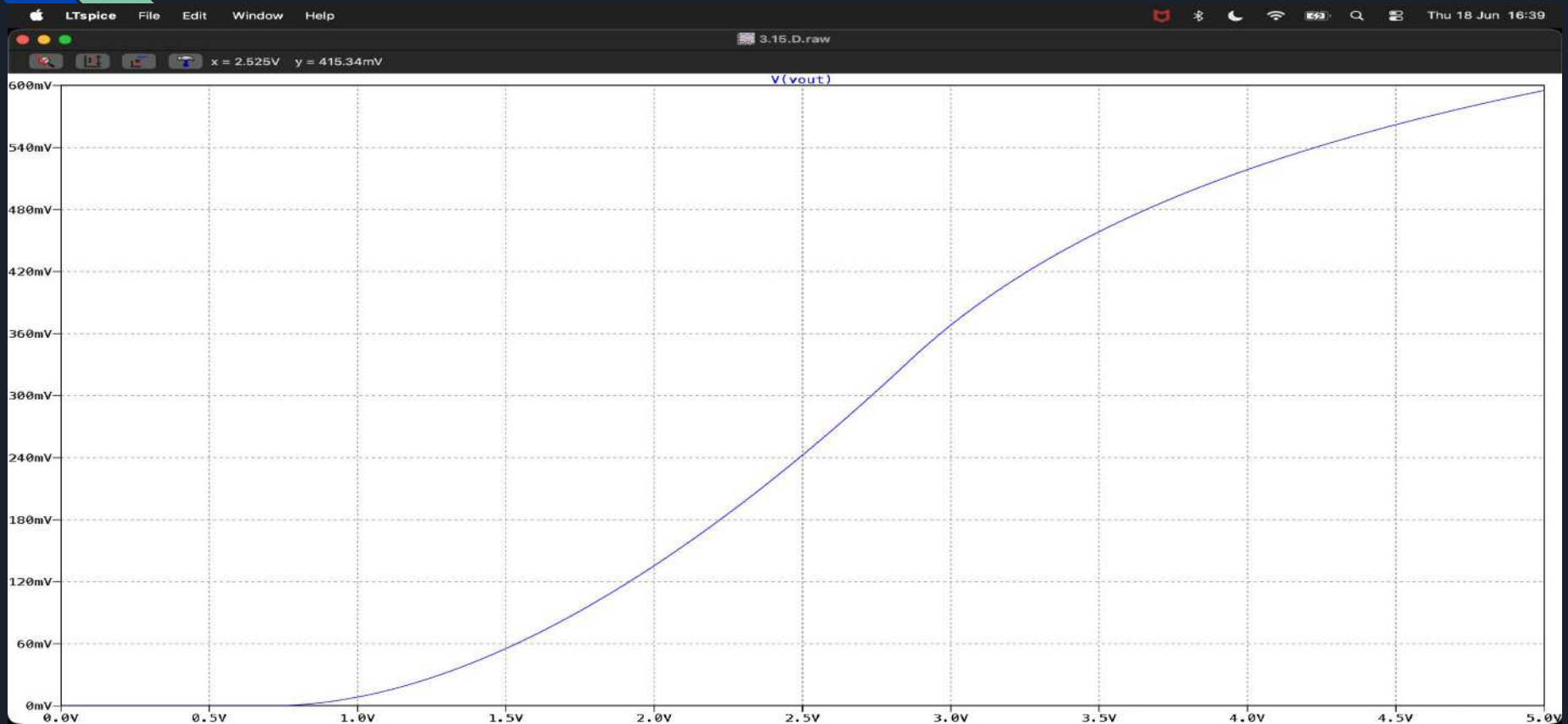
using ① & ② we can find exact value of V_{in}



LTSPICE Schematic



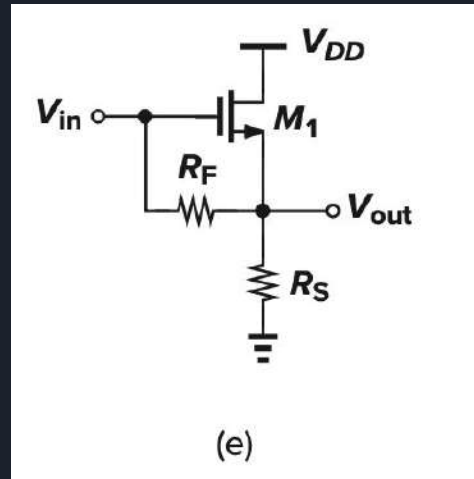
Simulation Results for Vout



Problem 3.15-e

3.15. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.78 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

Diagram -



Handwritten Solution

3.15/28

Before MOSFET turns ON
i.e. $V_{GS} < V_{th}$

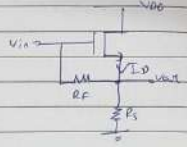
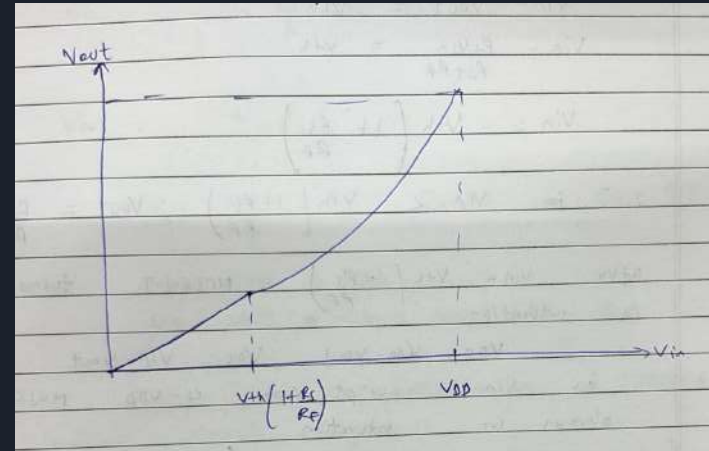
$V_{out} = \frac{R_S V_{in}}{R_S + R_F}$

MOSFET turns ON at $V_{GS} = V_{th}$

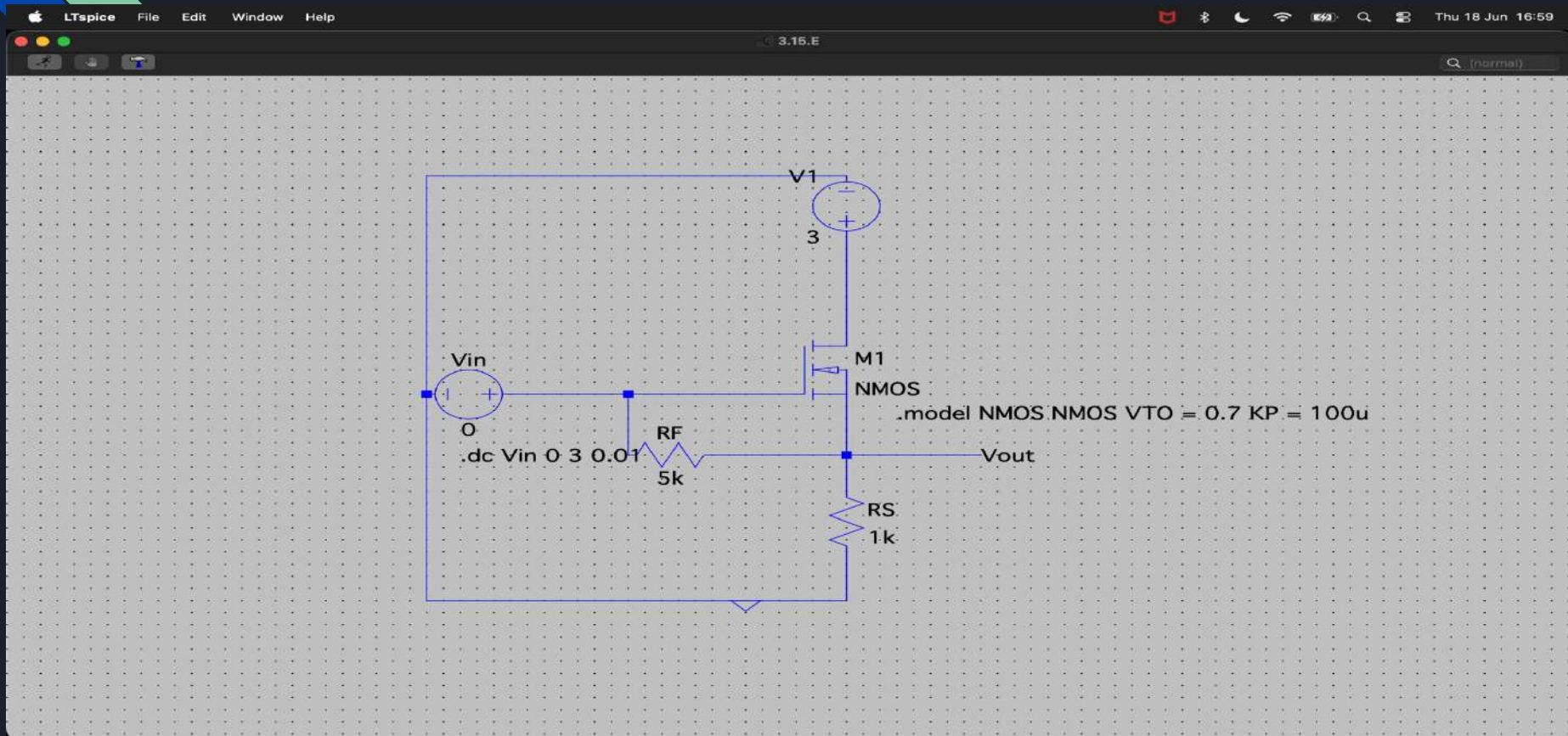
$V_{in} - V_{out} = V_{th}$
 $V_{in} - \frac{R_S V_{in}}{R_S + R_F} = V_{th}$
 $V_{in} = \frac{V_{th} \left(1 + \frac{R_S}{R_F} \right)}{1}$

$\therefore$ for $V_{in} < \frac{V_{th} \left(1 + \frac{R_S}{R_F} \right)}{1}$, $V_{out} = \frac{R_S V_{in}}{R_S + R_F}$

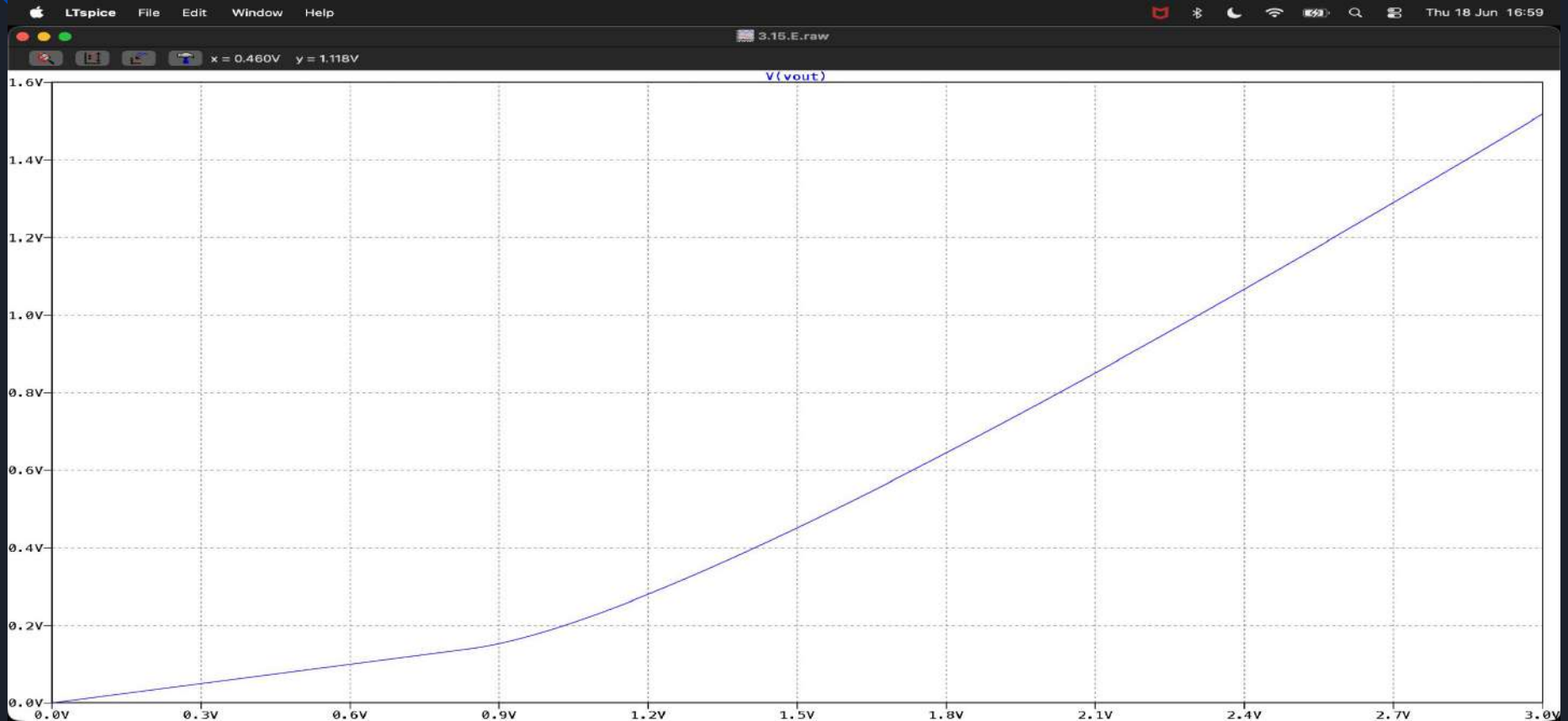
after $V_{in} = \frac{V_{th} \left(1 + \frac{R_S}{R_F} \right)}{1}$, MOSFET turns ON
 in saturation
 $V_{GS} = V_{GS} - V_{out}$, $V_{GS} = V_{in} - V_{out}$
 as V_{in} is swept from 0 to V_{DD} , MOSFET is
 always in saturation
 $I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left(V_{in} - V_{out} - V_{th} \right)^2$
 $I_{RF} = \frac{V_{in} - V_{out}}{R_F}$
 $I_{DS} = I_D + I_{RF}$
 $V_{out} = I_{DS} \cdot R_S$

LTSPICE Schematic



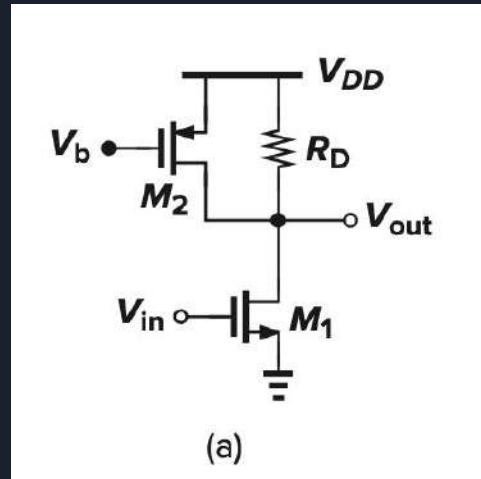
Simulation Results for Vout



Problem 3.16-a

3.16. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.79 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

Diagram -



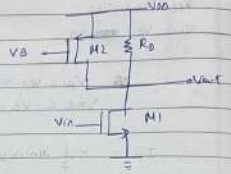
Handwritten Solution

3.67a

When $V_{in} < V_{th1}$, $M1$ is OFF

$\therefore$ no current flows

$V_{out} = V_{DD}$



As $V_{in} > V_{th1}$, current starts to flow
 $M1$ turns on in saturation

Regions of operation

i) $M2$ triode $\therefore V_{out} > V_D - V_{tp2}$

ii) $M2$ saturation $\therefore V_{out} < V_D - V_{tp2}$

iii) $M1$ triode $\therefore V_{out} < V_{in} - V_{th1}$

$M1$ saturation $\therefore V_{out} > V_{in} - V_{th1}$

As V_{in} increases I_D increases

By KCL at V_{out} node

$I_D = I_{D1} + I_{D2}$

$I_{D1} \uparrow, I_{D2} \uparrow, V_{out} \downarrow$ $V_{out} = V_{DD} - R_D I_D$

Region 1 $\therefore V_{in} < V_{th1}$

$I_1 = I_2 = I_D = 0$

$V_{out} = V_{DD}$

Region 2 $\therefore V_{in} > V_{th1}, V_{out} > V_D - V_{tp2}$ ($V_{DD} = 0.9V$)

$M1$ in saturation, $M2$ triode

$I_{D1} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_{in} - V_{th1})^2, I_{D2} = \mu_p C_{ox} \left(\frac{W}{L} \right)_2 \left[\frac{V_{DD} - V_D - V_{tp2}}{2} (V_{DD} - V_{tp2} - V_{out}) + \frac{(V_{DD} - V_{tp2})^2}{2} \right]$

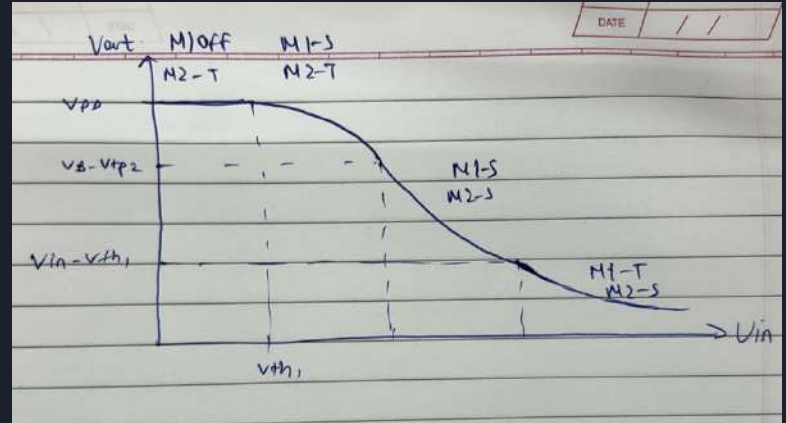
Region 3 $\therefore V_{in} > V_{th1}, V_{out} < V_D - V_{tp2}$

$M1$ and $M2$ in saturation

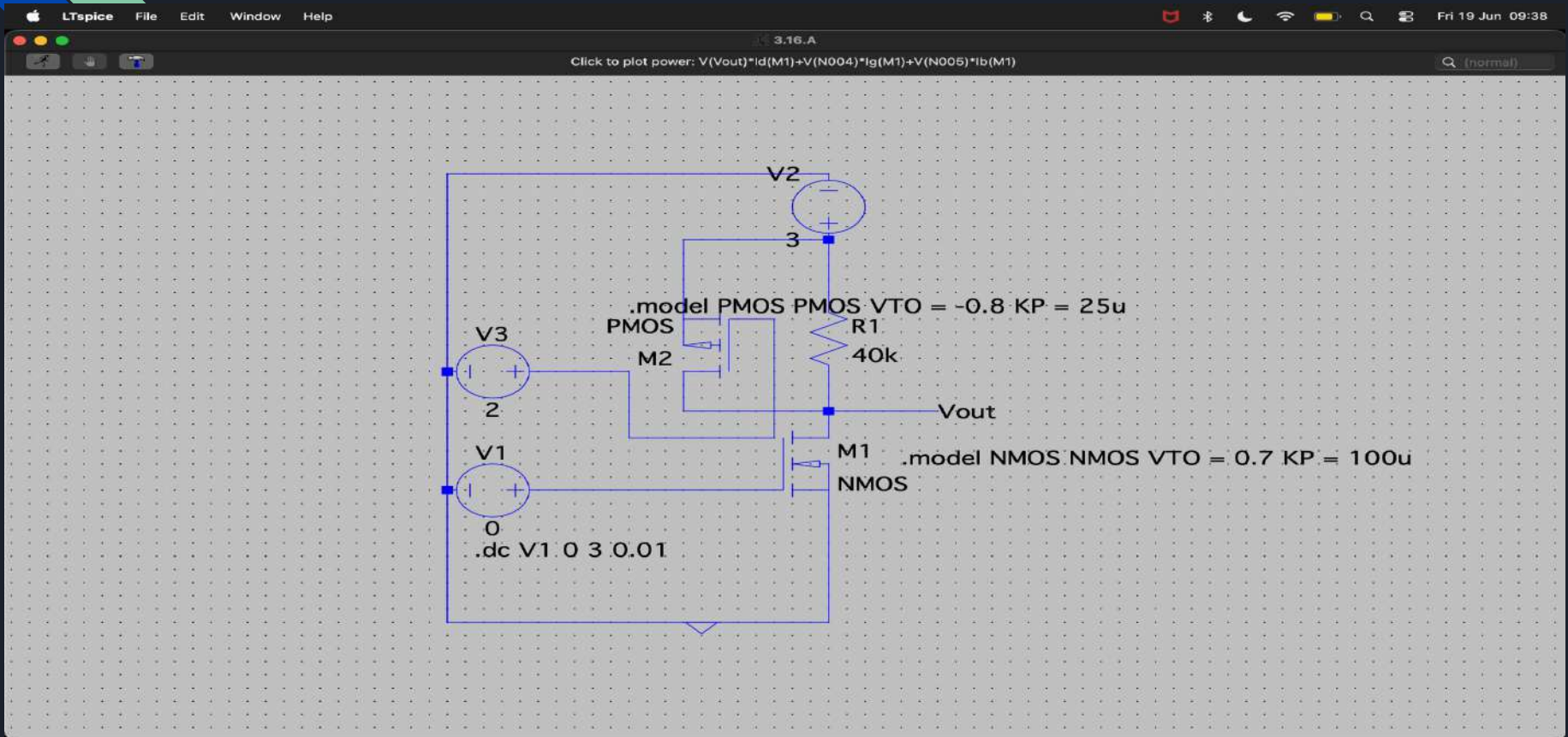
$V_{out} < V_{in} - V_{th1}$

Region 4 $\therefore V_{in} > V_{th1}, V_{out} < V_{in} - V_{th1}$

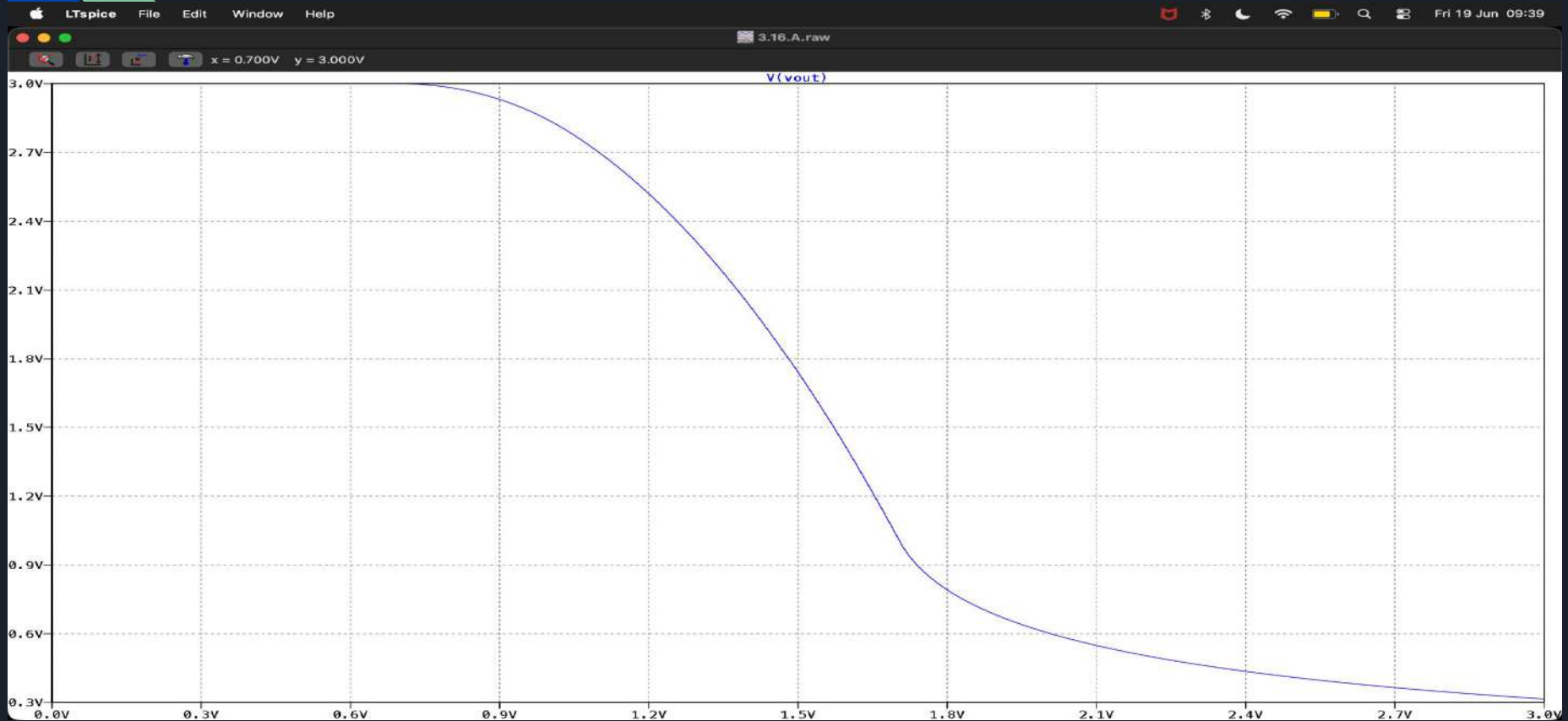
$M1 \rightarrow$ triode, $M2$ saturation



LTSPICE Schematic



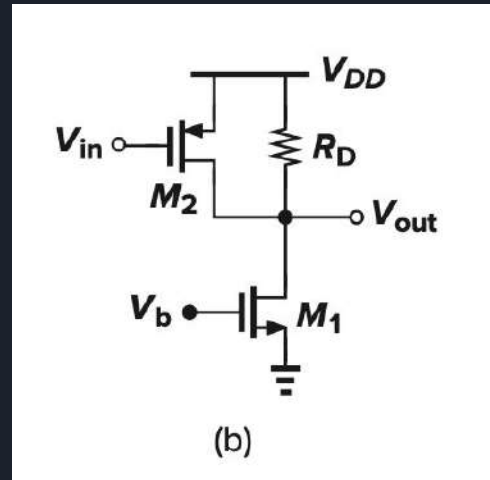
Simulation Results for Vout



Problem 3.16-b

3.16. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.79 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

Diagram -



Handwritten Solution

3.14.10b)

M2 turns OFF for $V_{GS2} > V_{tp2}$

$V_{in} - V_{DD} > V_{tp2}$
 $V_{in} > V_{DD} + V_{tp2}$

M2 Saturation - triode boundary

$V_{in} - V_{DD2} = V_{out}$

for $V_{out} > V_{in} - V_{tp2}$

$$I_2 = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} \left((V_{DD} - V_{in} - V_{tp}) (V_{DD} - V_{out}) - \frac{(V_{DD} - V_{out})^2}{2} \right)$$

as V_{in} increases I_2 decreases

at V_{out} node

$$I_1 = I_2 + I_R$$

$$I_R = I_1 - I_2$$

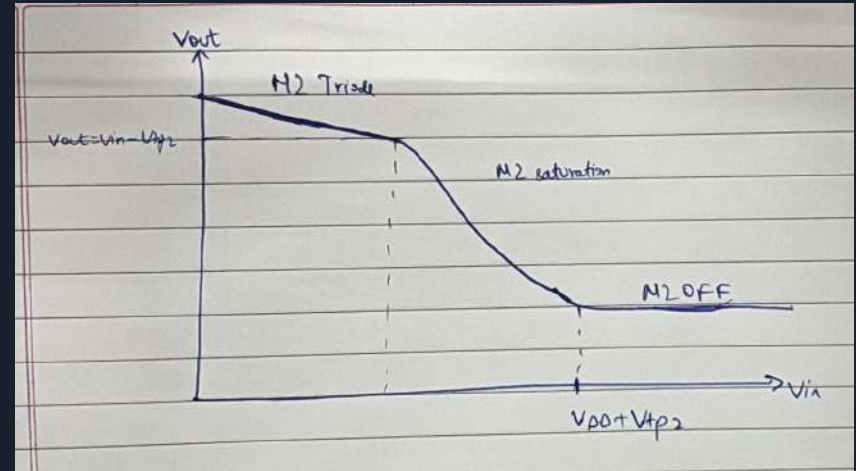
$V_{out} - V_{DD} = R_D I_R$

$$V_{out} = V_{DD} - R_D (I_1 - I_2)$$

$\therefore$ till $V_{in} = V_{DD} + V_{tp2}$
 as V_{in} increases V_{out} decreases

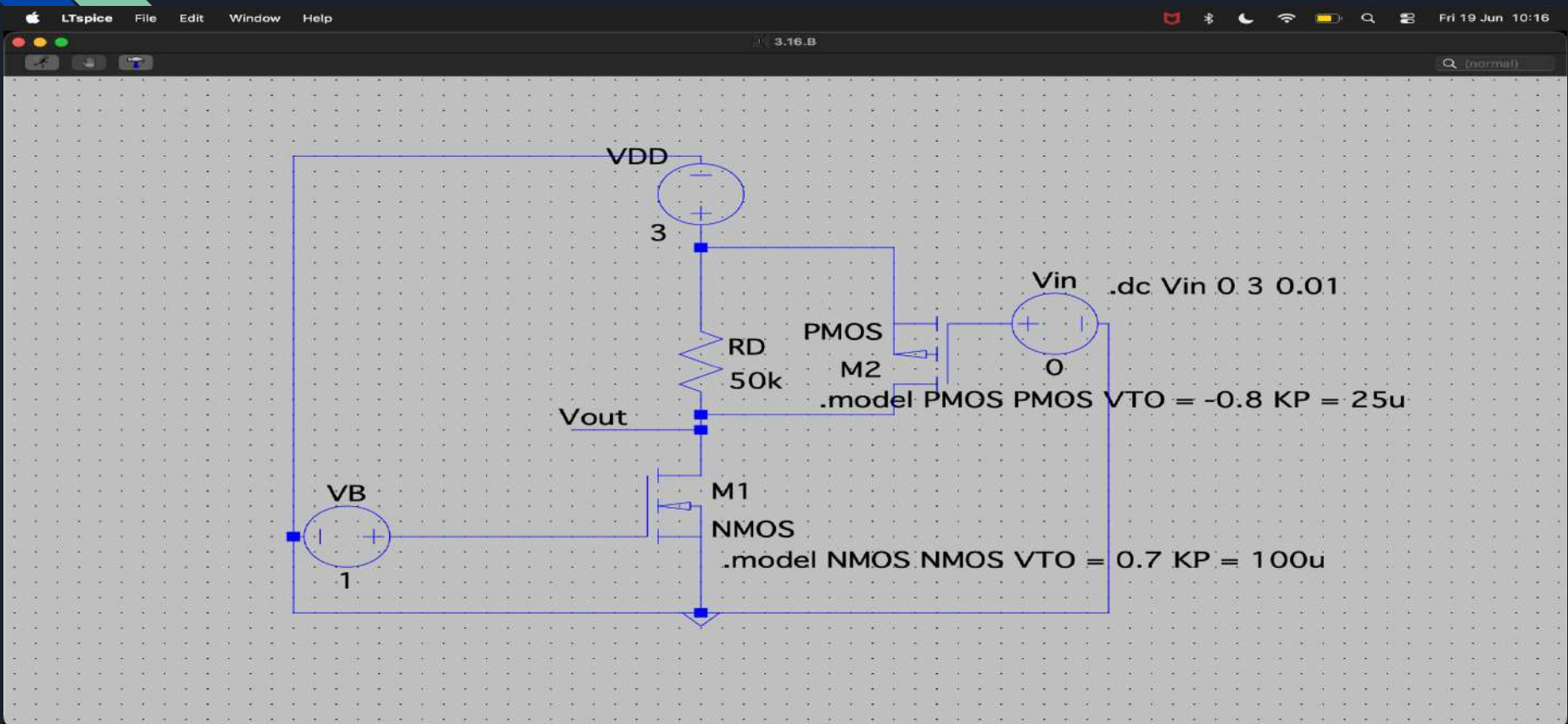
And after M2 turns OFF, current I_2 becomes 0, and $I_R = I_1 = \text{constant}$

$\therefore V_{out} = \text{constant} = V_{DD} - R_D I_1$

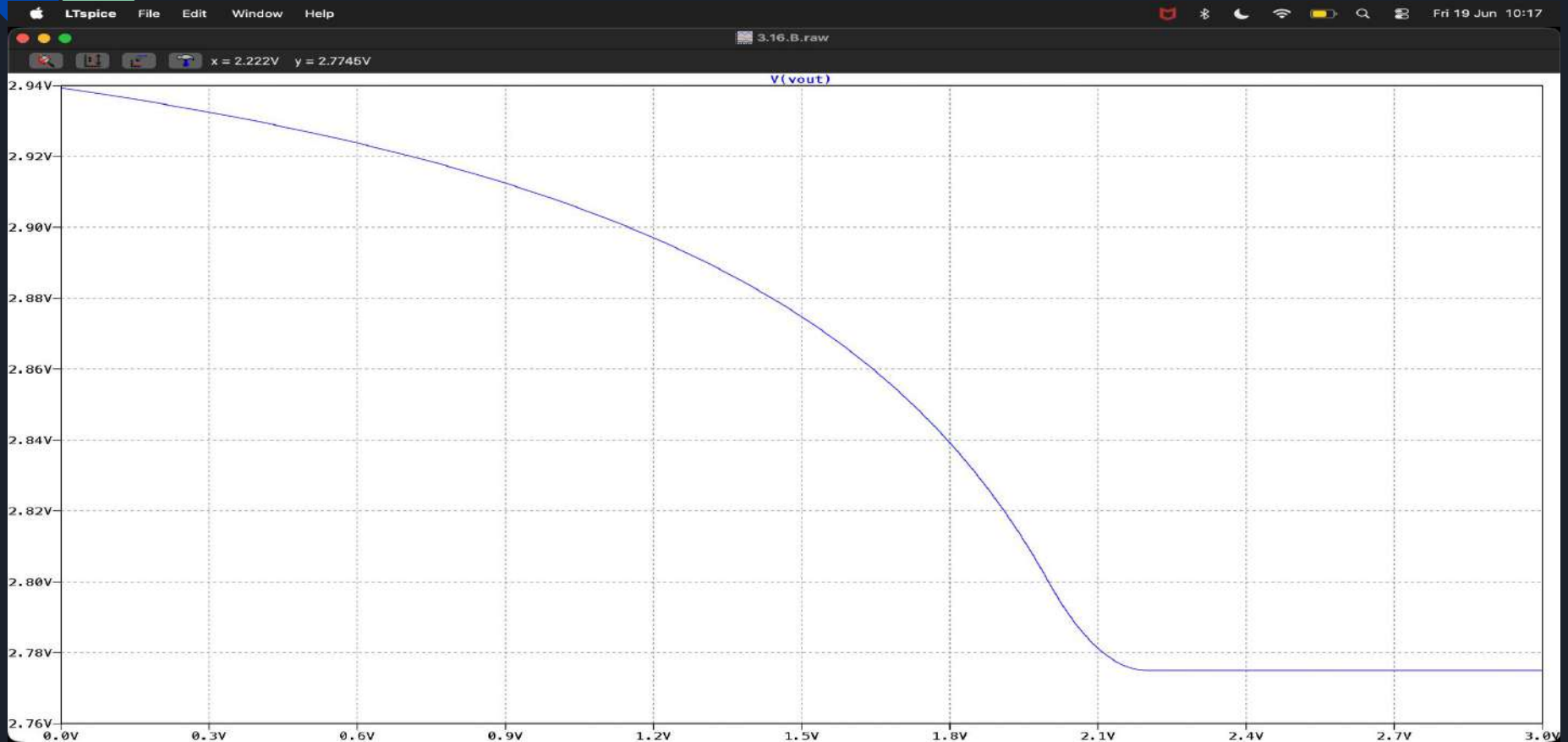


- M2 in triode: I_D increases linearly with $V_{in} \rightarrow$ smaller slope (gentle curve).
- M2 in saturation: I_D increases by square law with $V_{in} \rightarrow$ faster increase $\rightarrow$ steeper slope.

LTSPICE Schematic



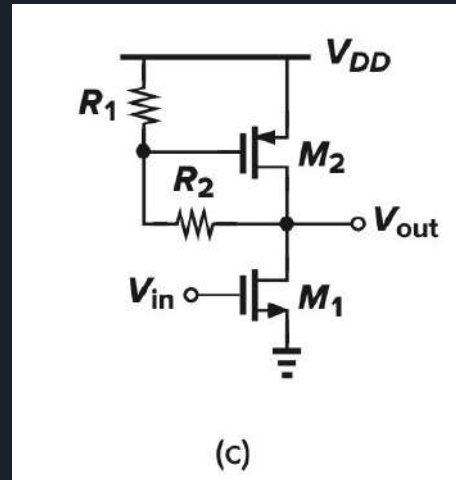
Simulation Results for Vout



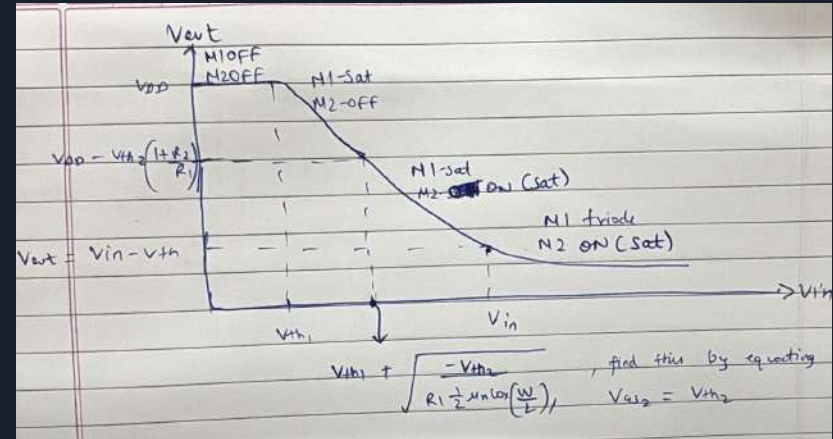
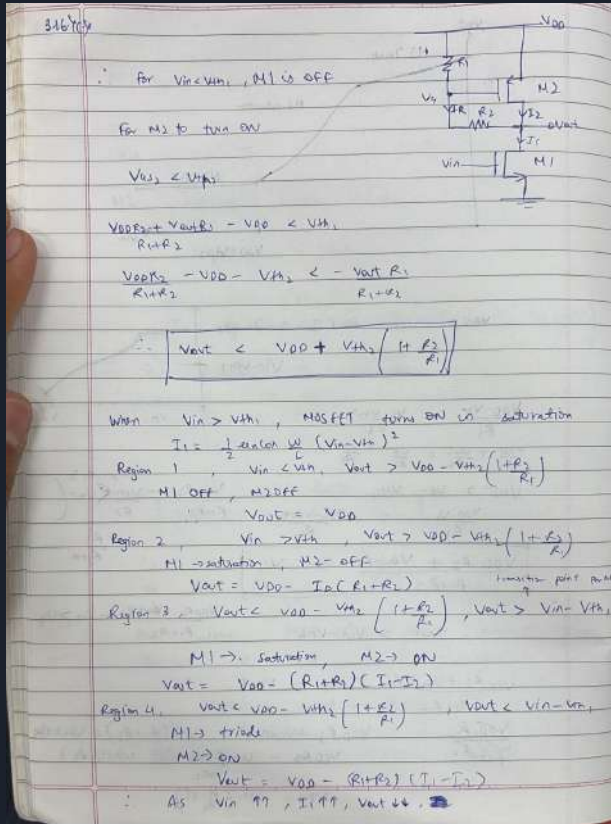
Problem 3.16-c

3.16. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.79 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

Diagram -

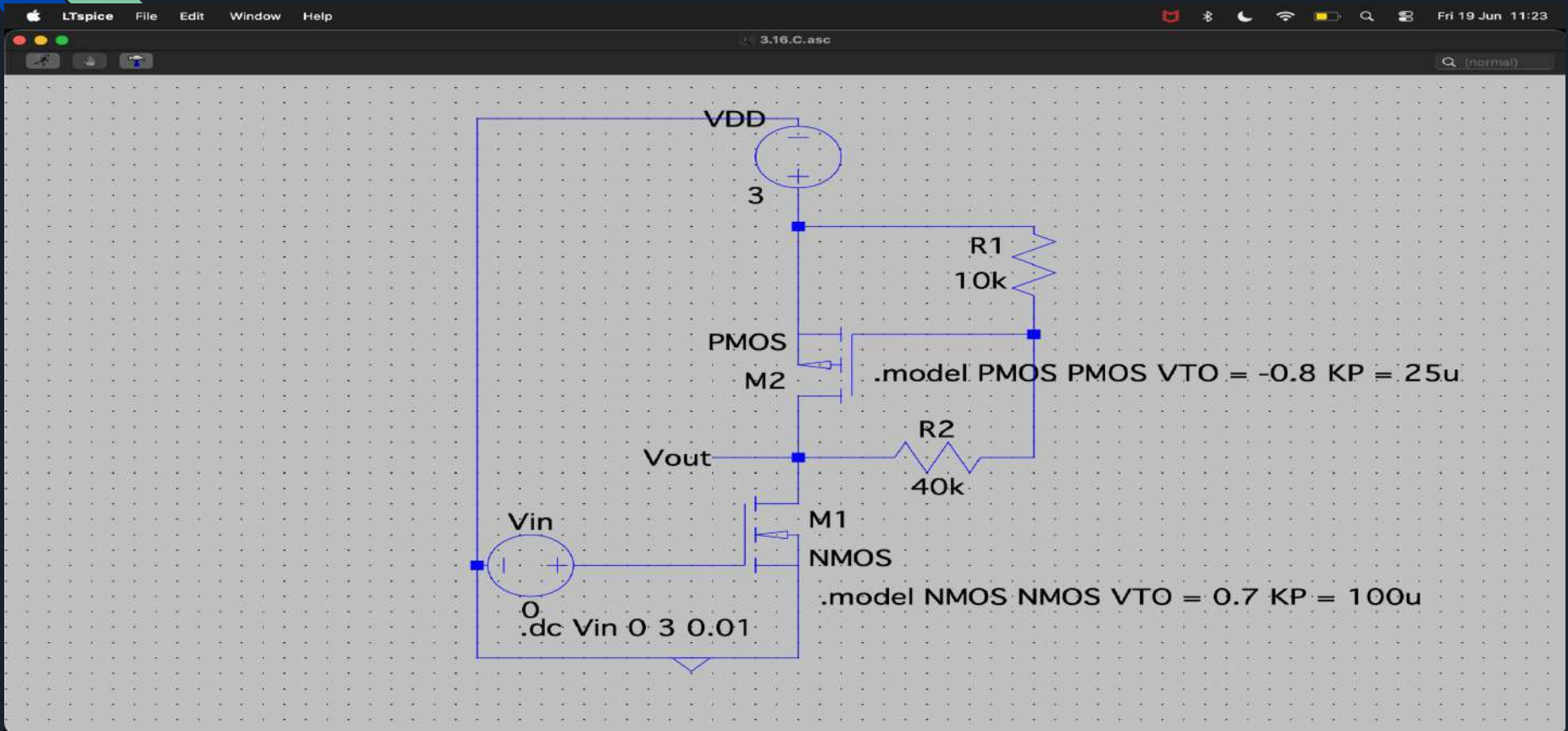


Handwritten Solution

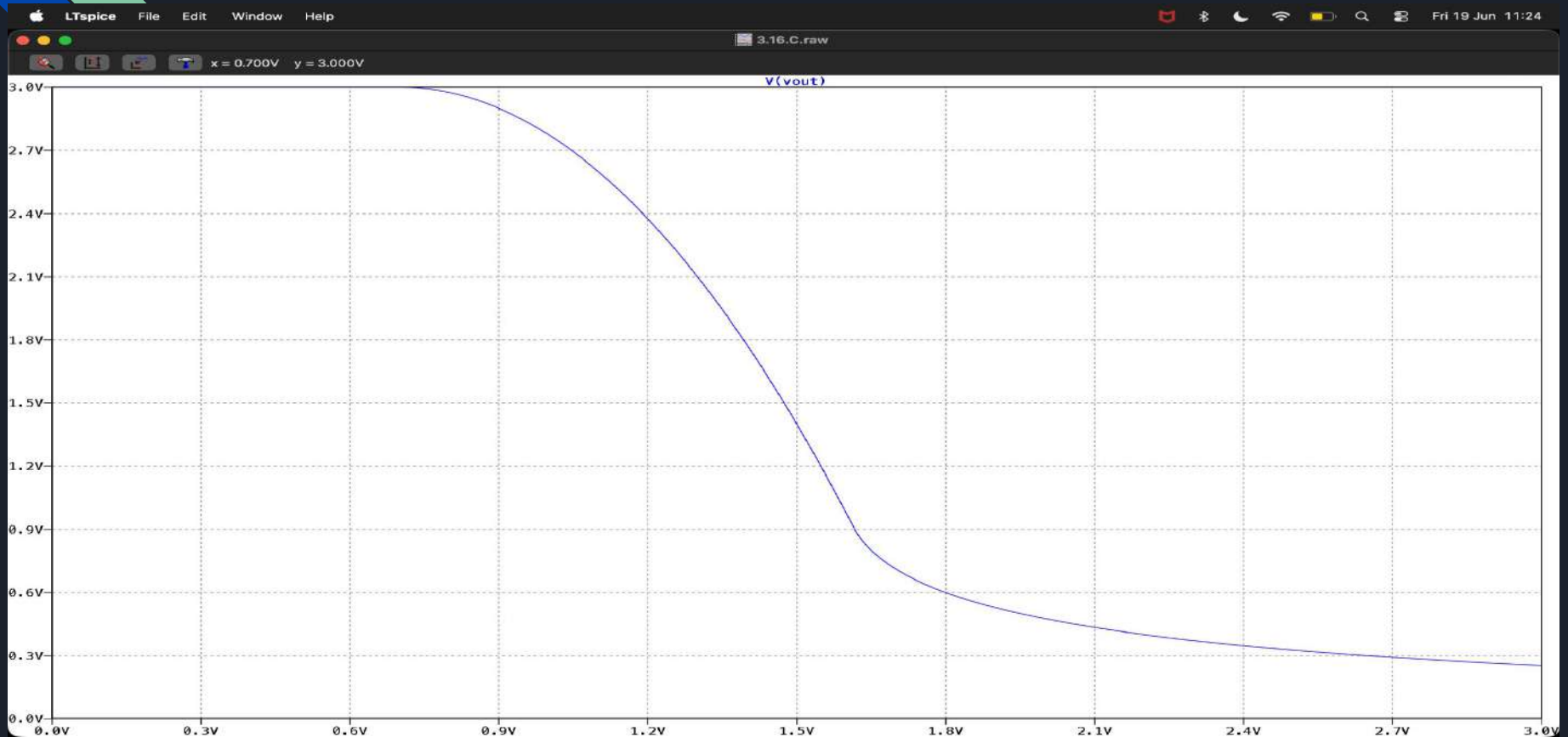


- M2 turns ON in saturation, and as V_{out} decreases, $V_{SD2} = V_{DD} - V_{out}$ increases, which keeps M2 more strongly in saturation.

LTSPICE Schematic



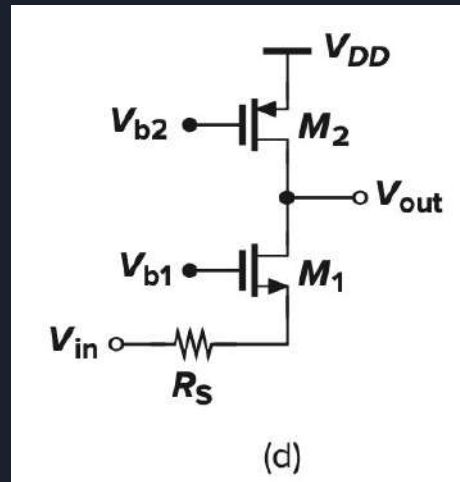
Simulation Results for Vout



Problem 3.16-d

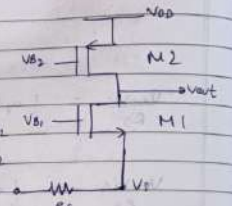
3.16. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.79 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

Diagram -



Handwritten Solution

316 Yd



Region	M1	M2
cut OFF	$V_{G1} = V_{in} < V_{th}$	X
Triode	$V_{out} < V_{G1} - V_{th1}$	$V_{out} > V_{G2} - V_{th2}$
Saturation	$V_{out} > V_{G1} - V_{th1}$	$V_{out} < V_{G2} - V_{th2}$

Assuming $V_{G1} < V_{G2}$

∴ Region 1, $V_{out} < V_{G1} - V_{th}$
M1 - Triode, M2 - saturation

∴ $V_{S1} = V_{in} + I_{D1} R_S$

As $V_{in} \uparrow$, $V_{S1} \uparrow$, ∴ $V_{GS1} \downarrow$, M1 strength reduces
∴ $V_{out} \uparrow$
when $V_{in} = 0$, $V_{S1} = I_{D1} R_S$

$I_D = I_{D1} = I_{D2} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_2 (V_{DD} - V_{th2} - (V_{out} - I_{D1} R_S))^2$

∴ $I_D = I_{D1} = I_{D2} = I_{D1}(\text{triode})$

uncouple $\left((V_{G1} - I_{D1} R_S - V_{th1})(V_{out} - I_{D1} R_S) - (V_{out} - I_{D1} R_S)^2 \right) = I_D$

∴ From here we can find V_{out} = fixed value

As $V_{in} \uparrow$, $V_{S1} \downarrow$, ∴ $V_{out} \uparrow$

Region 2, $V_{out} > V_{G1} - V_{th1}$, $V_{out} < V_{G2} - V_{th2}$
M1 → saturation, M2 → saturation

∴ As $V_{in} \uparrow$, I_D has to be constant ideally
∴ V_{out} gets pulled up, i.e. $V_{out} \uparrow$
Practically due to channel length modulation, I_D starts to increase, stabilising V_{out}

Region 3, $V_{out} > V_{G2} - V_{th2}$
M1 - saturation, M2 - triode

$I_{D1} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_{G1} - V_{in} - I_{D1} R_S - V_{th1})^2$

$I_{D2} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_2 \left((V_{DD} - V_{G1} - V_{th1})(V_{DD} - V_{out}) - (V_{DD} - V_{G1} - V_{th1})^2 \right)$

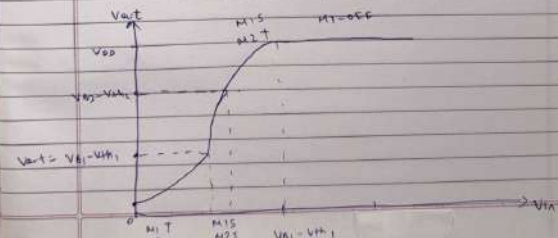
as V_{in} increases, I_{D1} ↓ ↓

$I_{D1} = I_{D2}$, I_{D1} also has to decrease
∴ $V_{DD} - V_{out}$ has to decrease
∴ $V_{out} \uparrow$

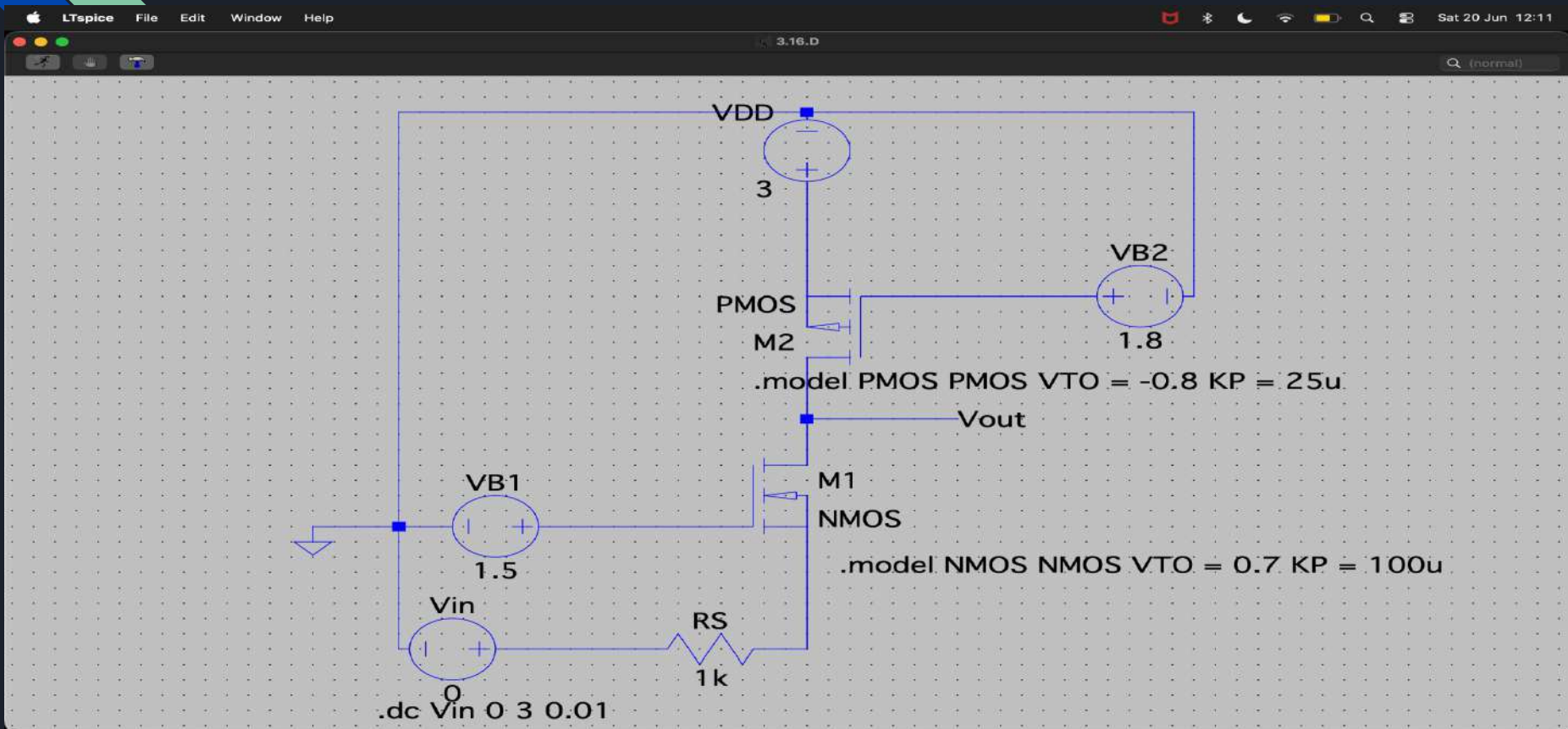
As $V_{in} \uparrow$, $V_{S1} = V_{in} + I_{D1} R_S$,
 I_{D1} becomes almost zero, ∴ $V_{S1} \approx V_{in}$

∴ $V_{G1} = V_{G2} = V_{in}$

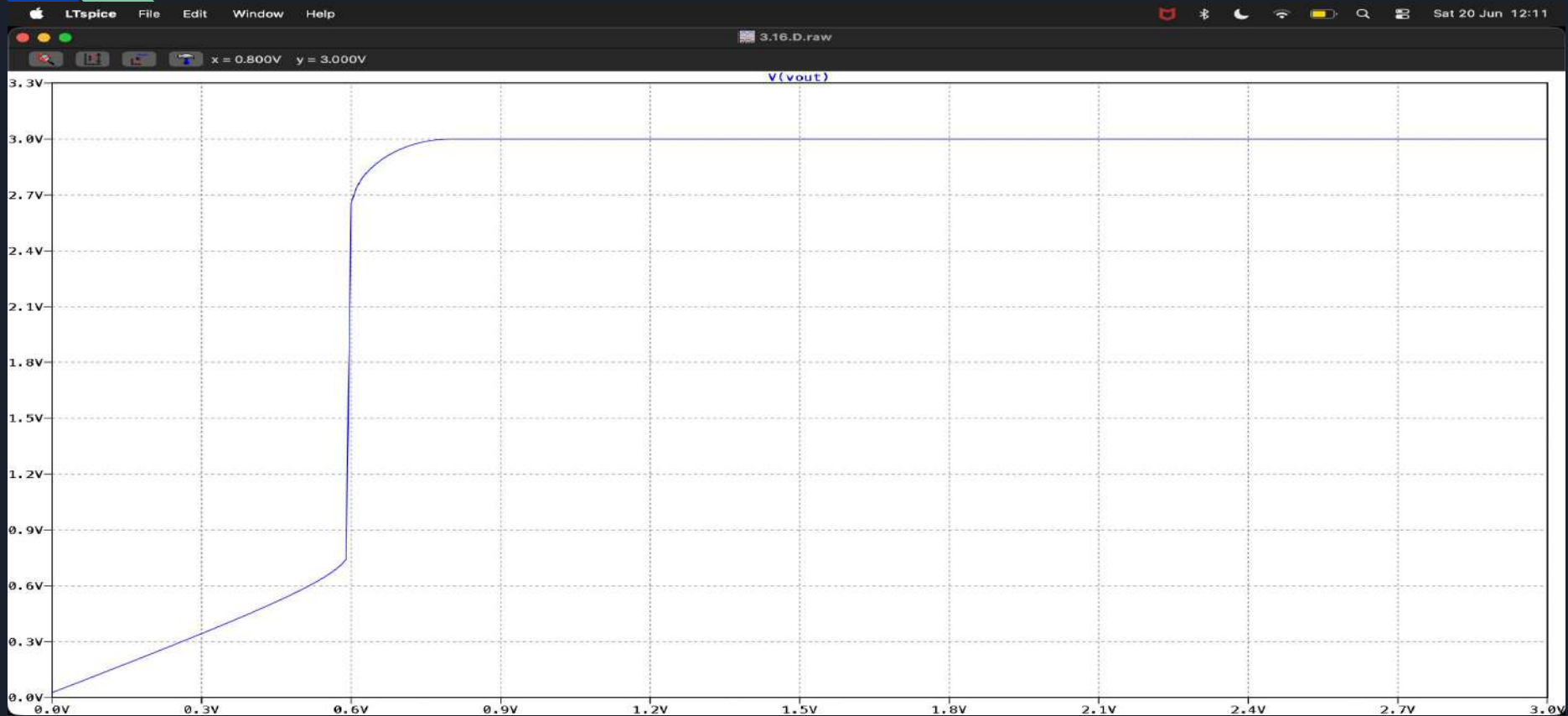
∴ M1 will turn OFF when
 $V_{GS1} < V_{th1}$
 $V_{G1} - V_{in} < V_{th1}$ (roughly)
∴ $V_{in} > V_{G1} - V_{th1}$
M1 turns OFF, ∴ $V_{out} = V_{DD}$



LTSPICE Schematic



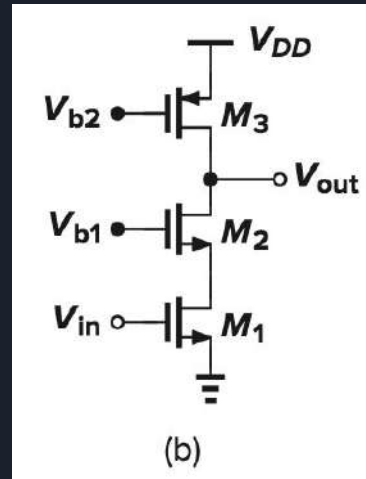
Simulation Results for Vout



Problem 3.17-b

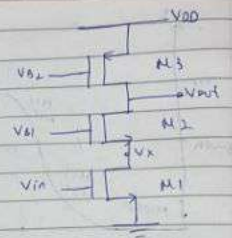
3.17. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.80 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

Diagram -



Handwritten Solution

317761



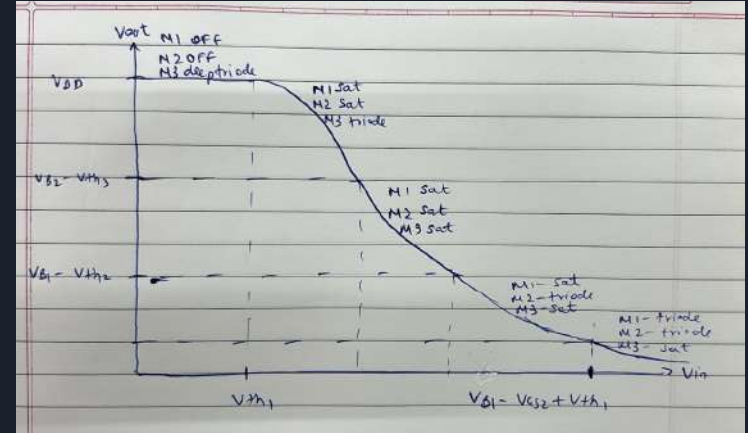
Let us consider the following table for the MOSFET operating regions.

Region	M1	M2	M3
Cut off	$V_{in} < V_{th1}$	$V_{d1} - V_x < V_{th2}$	$V_{d2} - V_{out} > V_{th3}$
Triode	$V_x < V_{in} - V_{th1}$	$V_{out} < V_{d1} - V_{th2}$	$V_{d2} - V_{th3} < V_{out}$
Saturation	$V_x > V_{in} - V_{th1}$	$V_{out} > V_{d1} - V_{th2}$	$V_{d2} - V_{th3} > V_{out}$

∴ for $V_{in} = V_{th1}$, MOSFET M1 is OFF
 $V_{out} = V_{DD}$

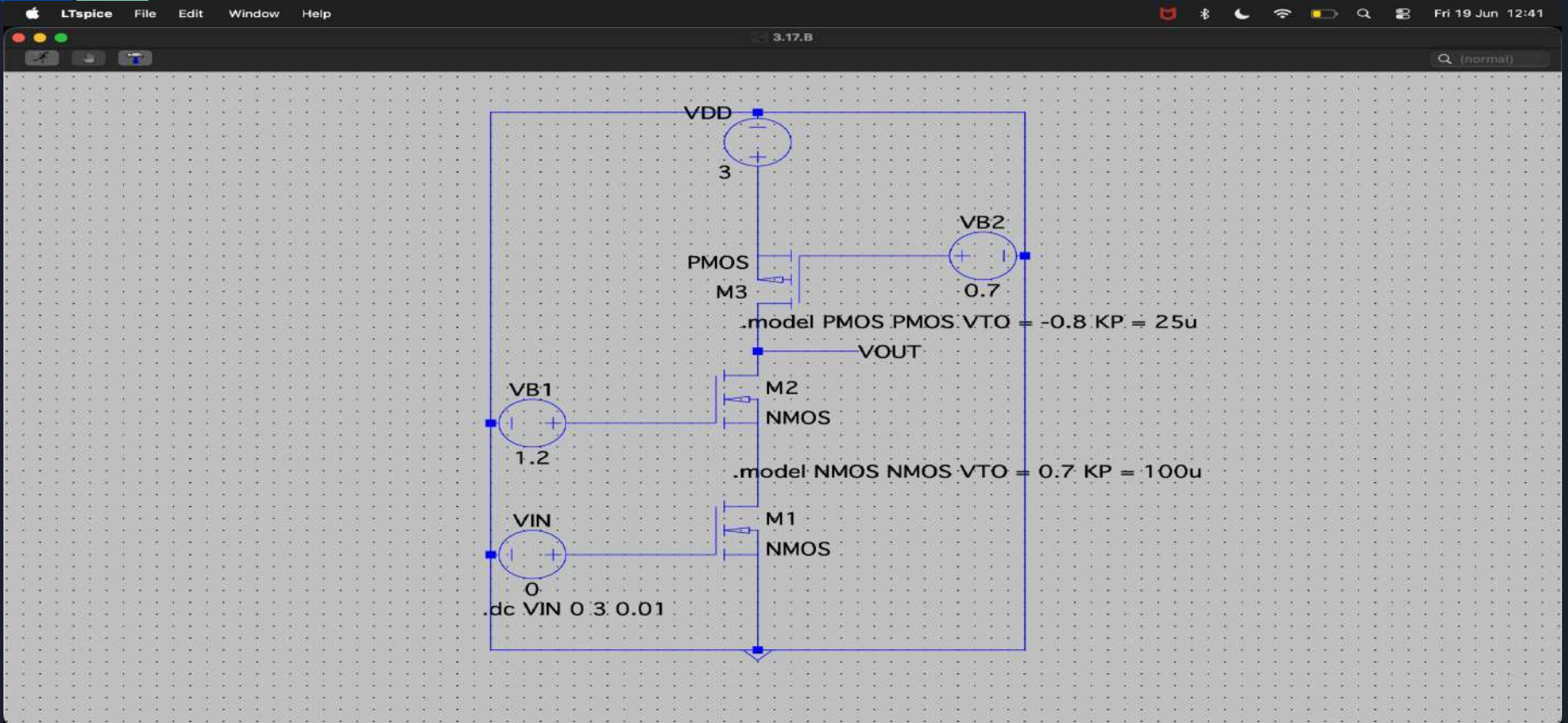
After $V_{in} = V_{th1}$, as V_{in} ↑↑, I ↑↑, V_{out} ↓↓

For M1 transition point, $V_x = V_{in} - V_{th1}$
 M2 is in saturation. ∴ $V_{d2} = V_{d1} - V_x$
 $V_x = -V_{gs2} + V_{g1}$
 $V_{in} = V_x + V_{th1}$
 $= V_{d1} - V_{gs2} + V_{th1}$

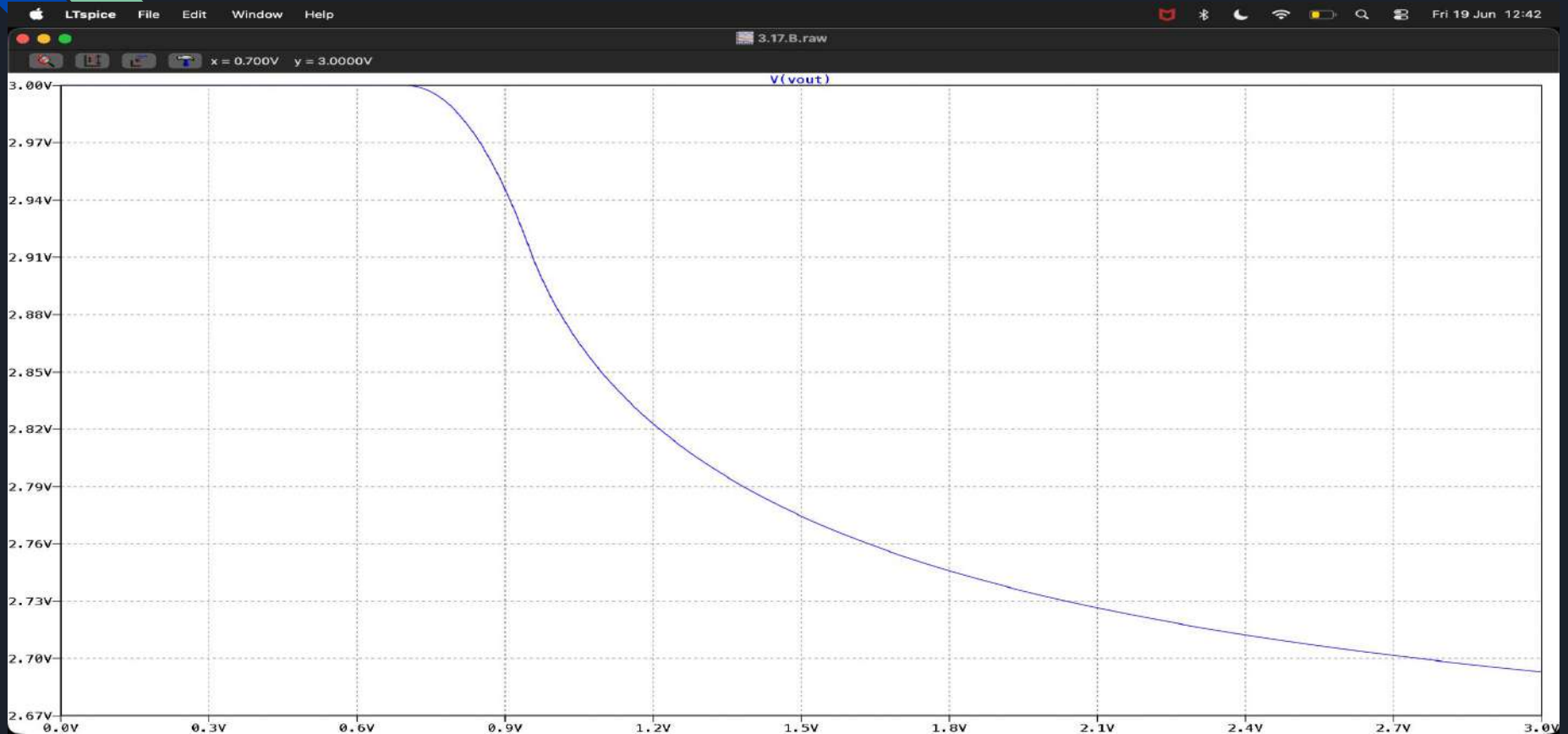


- When M1 is OFF, no current flows through the stack, so V_x rises close to V_{DD} . This reduces $V_{GS2} = V_{B2} - V_x$ below V_{TH2} , turning M2 OFF. Hence, for $V_{in} < V_{TH1}$, both M1 and M2 remain OFF.

LTSPICE Schematic



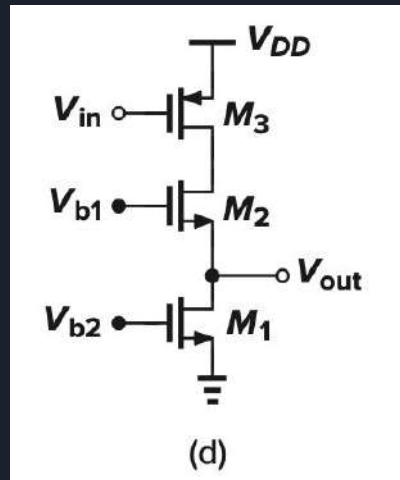
Simulation Results for Vout



Problem 3.17-d

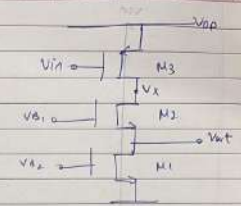
3.17. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.80 as V_{in} varies from 0 to V_{DD} . Identify important transition points.

Diagram -



Handwritten Solution

3.17.147



$\therefore M3$ turns OFF when $V_{in} - V_{DD} > V_{th3}$

$V_{in} > V_{DD} + V_{th3}$

Initially when $V_{in} = 0$, $V_{out} = +3V$
 I_s is very large.

But M1 gate is connected to V_{out}
 $V_{gs1} = V_{out}$
 $\therefore$ current is limited through M1

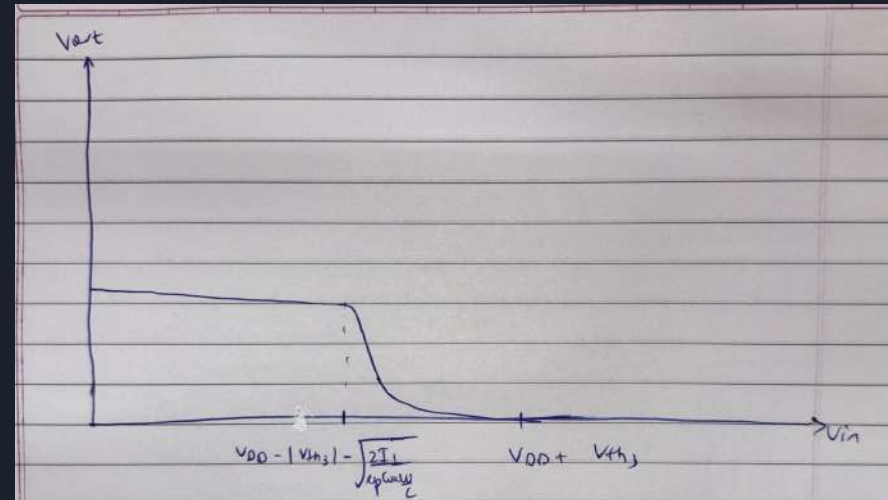
Due to this the voltage V_x rises close to V_{DD} and V_{out} also gets pushed high, putting M1 in saturation.

Due to this till I_s is greater than I_{sat} V_{out} remains almost constant.
 $I_s = I_{sat}$
 $\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_1 (V_{gs1} - V_{th1})^2 = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L}\right)_3 (V_{DD} - V_{in} - V_{th3})^2$

I_{sat}

$\therefore V_{in} = V_{DD} - V_{th3} - \sqrt{\frac{2 I_{sat}}{\mu_p C_{ox} \left(\frac{W}{L}\right)_3}}$

After this point if $V_{in} \uparrow$, $I_s \downarrow$, $\therefore I_s$ has to decrease and M1 enters triode region, $\therefore V_{out}$ decreases

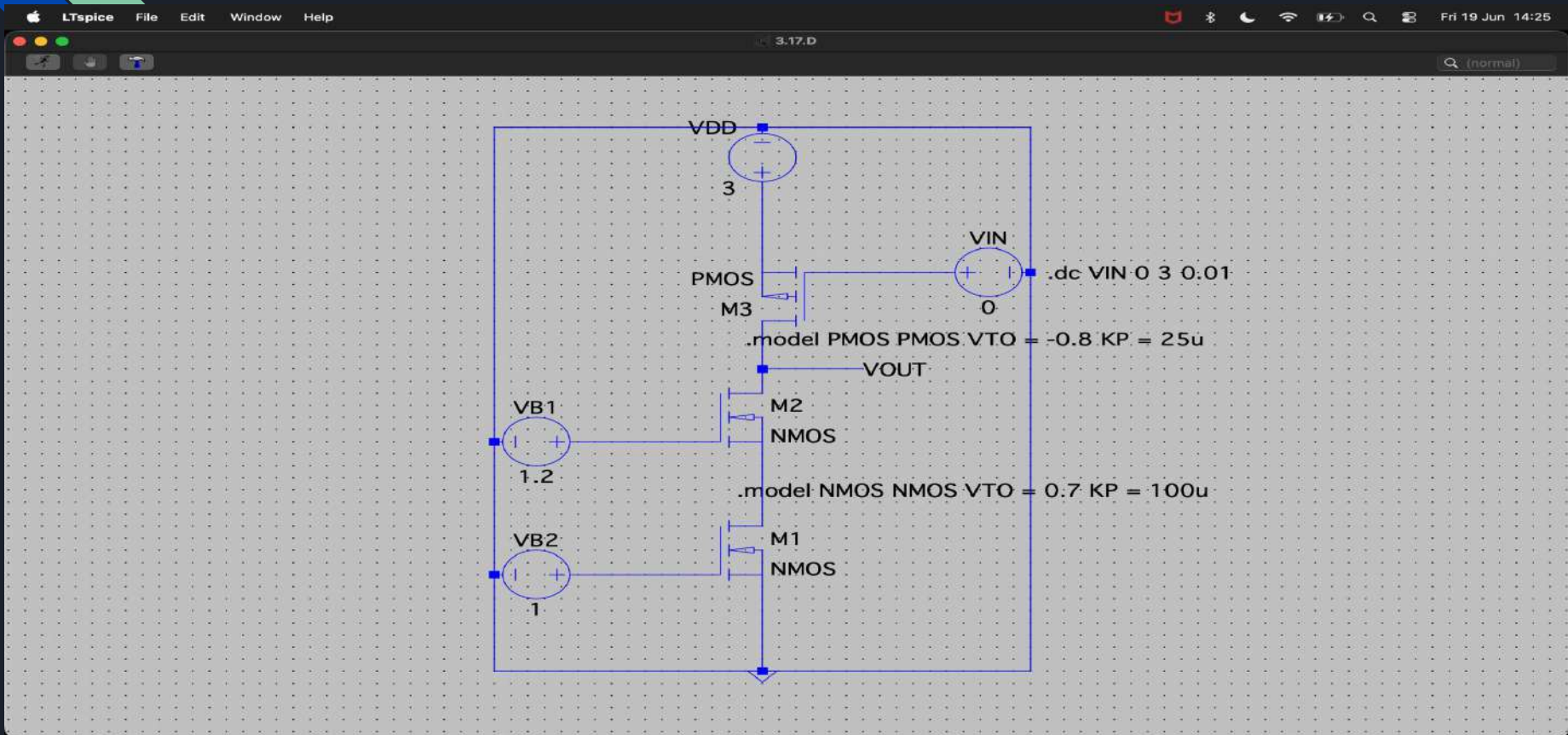




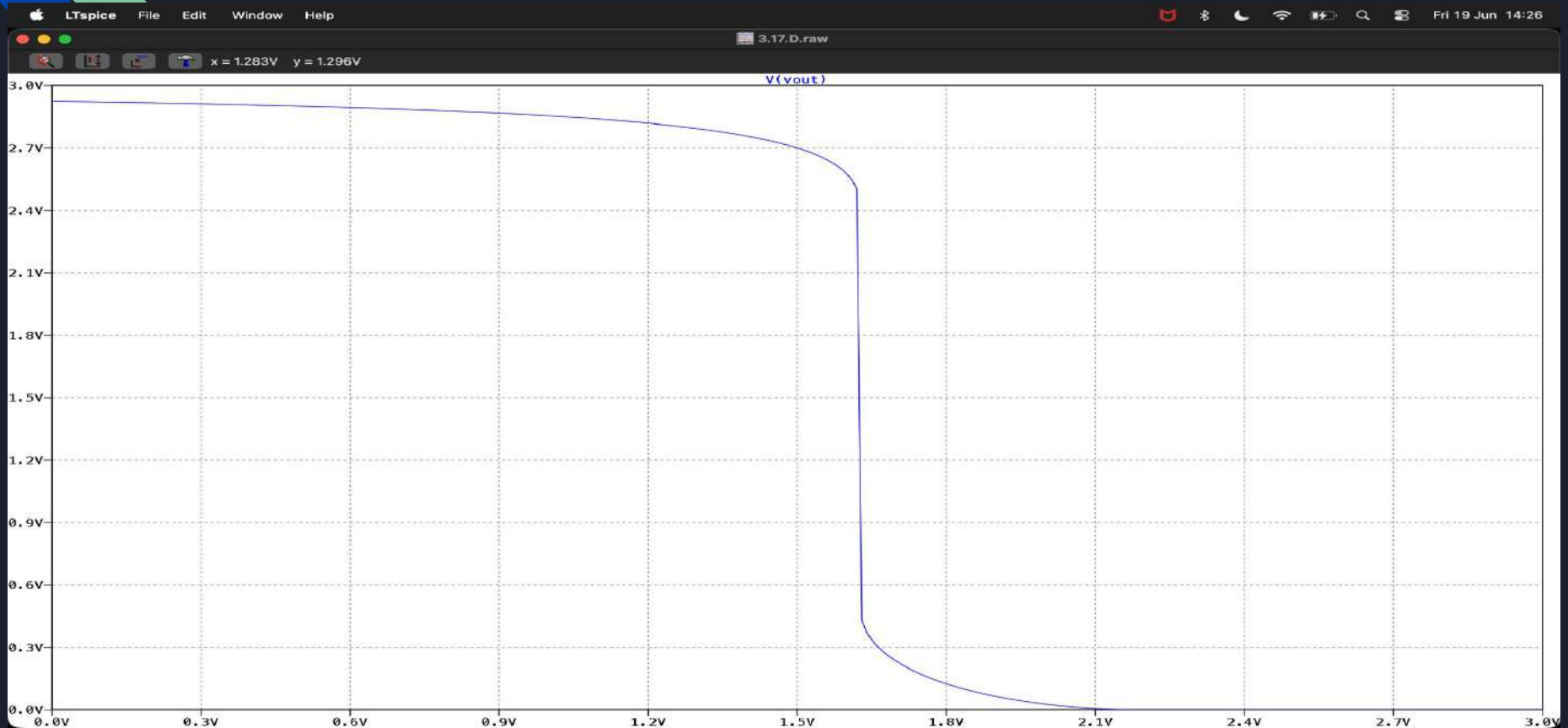
Observations

- **Region 1:** With $V_{in} = 0$, M3 is strongly ON and in triode due to low V_{SD3} ; it supplies excess current, raising node voltages and keeping M1 and M2 in saturation.
- **Region 2:** As V_{in} increases, M3 enters saturation due to channel pinch-off and limits the branch current; M1 becomes current-starved, reducing V_{DS1} and entering triode, while M2 remains in saturation due to sufficient drain voltage.

LTSPICE Schematic



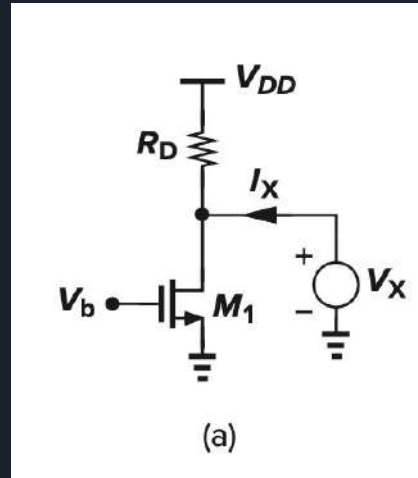
Simulation Results for Vout



Problem 3.18-a

3.18. Sketch I_X versus V_X for the circuits of Fig. 3.81 as V_X varies from 0 to V_{DD} . Identify important transition points.

Diagram -



Handwritten Solution

$\therefore$ When $V_X = 0$
 $I_D = 0$

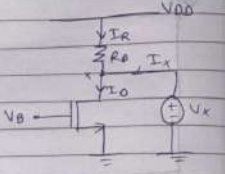
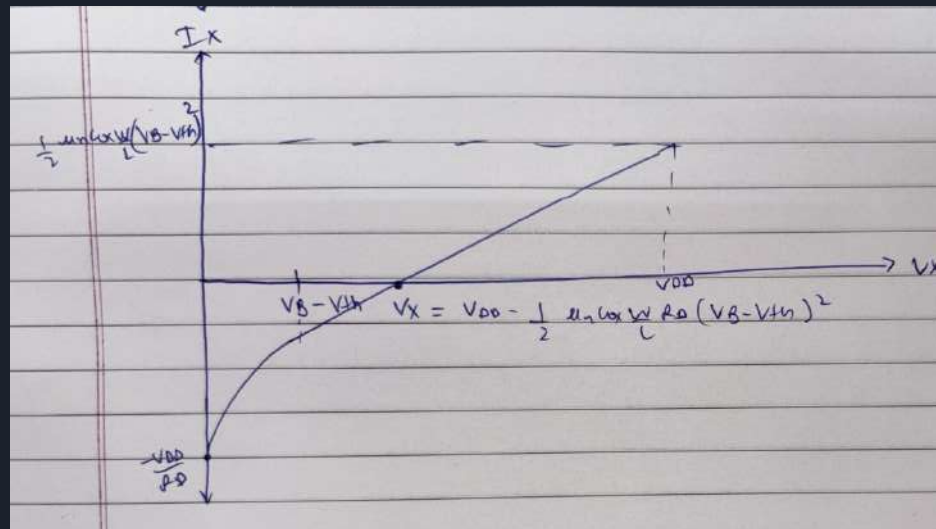
$I_X = -I_D = -\frac{V_{DD}}{R_D}$

$\therefore$ at node X
 $I_D = I_X + I_E$
 $I_X = I_D - I_E$

$\therefore$ When MOSFET is in triode region
 $V_X < V_B - V_{th}$
 $I_X = \mu_n C_{ox} \frac{W}{L} \left((V_B - V_{th}) V_X - \frac{V_X^2}{2} \right) - \left(\frac{V_{DD} - V_X}{R_D} \right)$

$\therefore$ When MOSFET is in saturation
 $V_X > V_B - V_{th}$
 $I_X = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_B - V_{th})^2 - \left(\frac{V_{DD} - V_X}{R_D} \right)$

We will get $I_X = 0$ at
 $I_D = I_E$
 $\frac{V_{DD} - V_X}{R_D} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_B - V_{th})^2$
 $V_X = V_{DD} - \frac{1}{2} \mu_n C_{ox} \frac{W}{L} R_D (V_B - V_{th})^2$

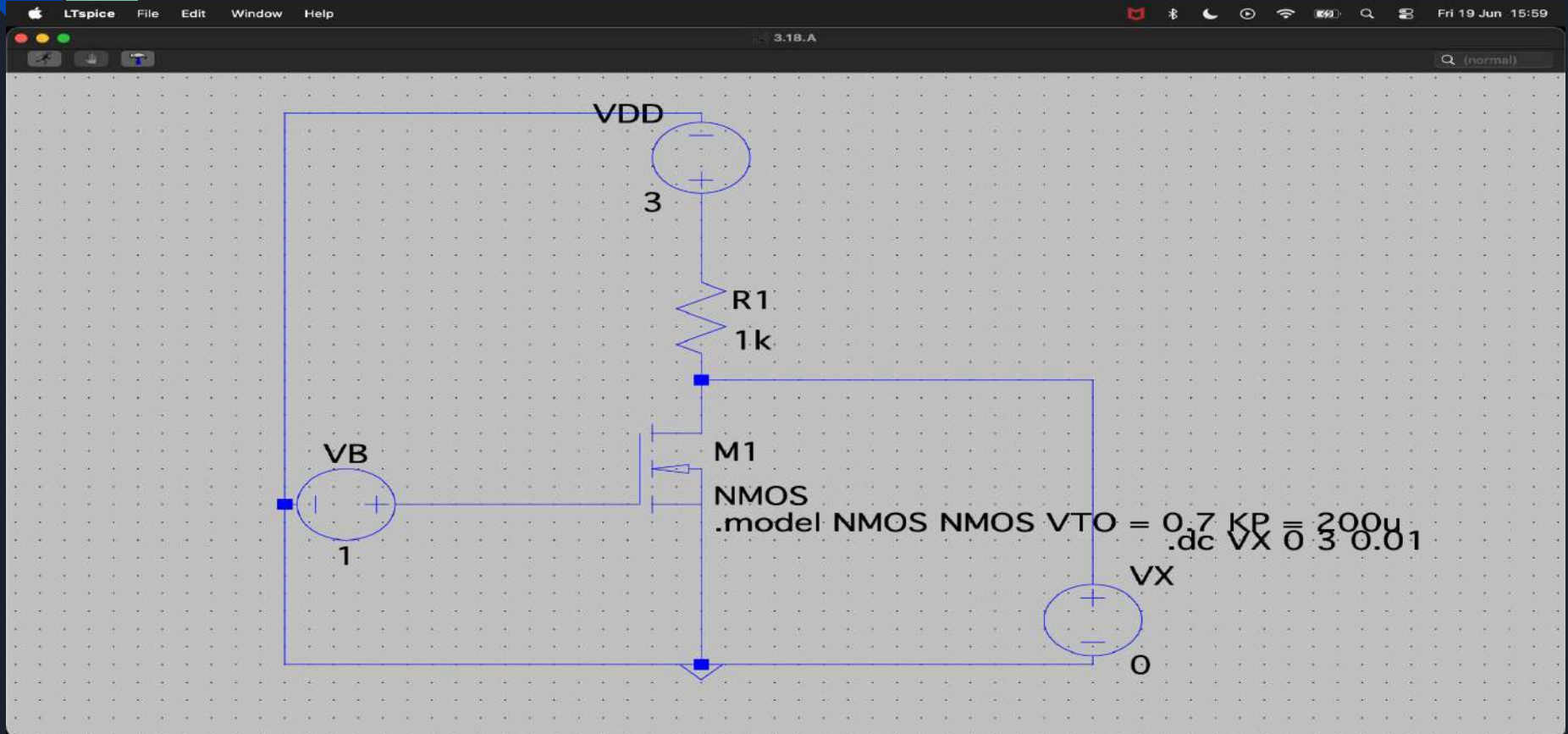





Points to be noted

- Initially MOSFET current $I_D = 0$, $I_X = -V_{DD}/R_D$.
- The I_X curve crosses the x-axis after the transition point because increasing V_X reduces the resistor current until it matches the transistor saturation current. Before transition, low V_X causes high resistor current, making I_X negative. At $V_X = V_b - V_{th}$, the transistor enters saturation and limits its current to I_{sat} , further increase in V_X reduces resistor current until $I_X = 0$.

LTSPICE Schematic



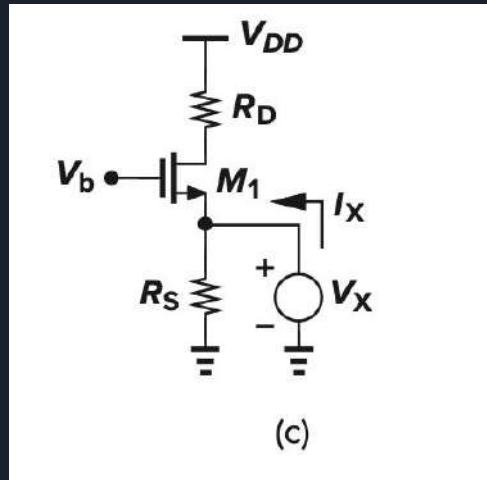
Simulation Results for I_x



Problem 3.18-c

3.18. Sketch I_X versus V_X for the circuits of Fig. 3.81 as V_X varies from 0 to V_{DD} . Identify important transition points.

Diagram -

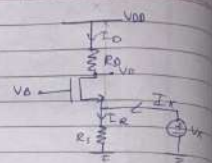


Handwritten Solution

12/11/21

MOSFET will turn OFF
when $V_{GS} < V_{th}$

$V_G - V_X \geq V_{th}$
 $V_X > V_G - V_{th}$



Initially, when $V_X = 0$
 $I_D = 0$
 $\therefore I_D = -I_S$
 When $V_X = 0$, $V_{GS} = V_G$, so V_X increases
 V_{GS} decreases
 Initially MOSFET has a wide (strong) channel
 I_D is high, V_D is low; initially MOSFET is in triode region
 For triode-saturation transition

$V_{DD} - I_D R_D = V_G - V_{th}$ ($V_{GS} = V_G - V_X$)
 $I_D = \frac{V_{DD} - V_G + V_{th}}{R_D}$

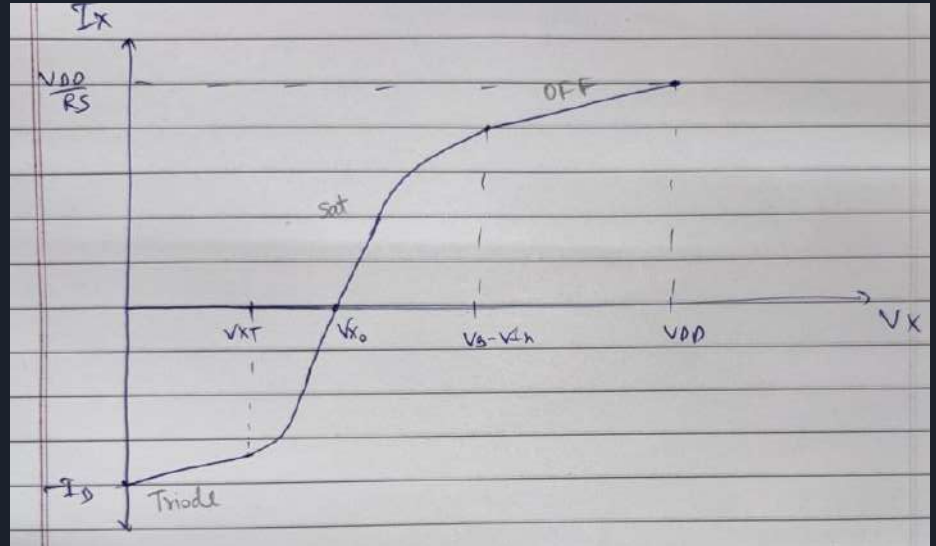
At this transition we can use Jansz equation
 $I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_G - V_X - V_{th})^2$

$V_X = V_G - V_{th} - \sqrt{\frac{2(V_{DD} - V_G + V_{th})}{\mu_n C_{ox} \frac{W}{L} R_D}} - V_{th}$

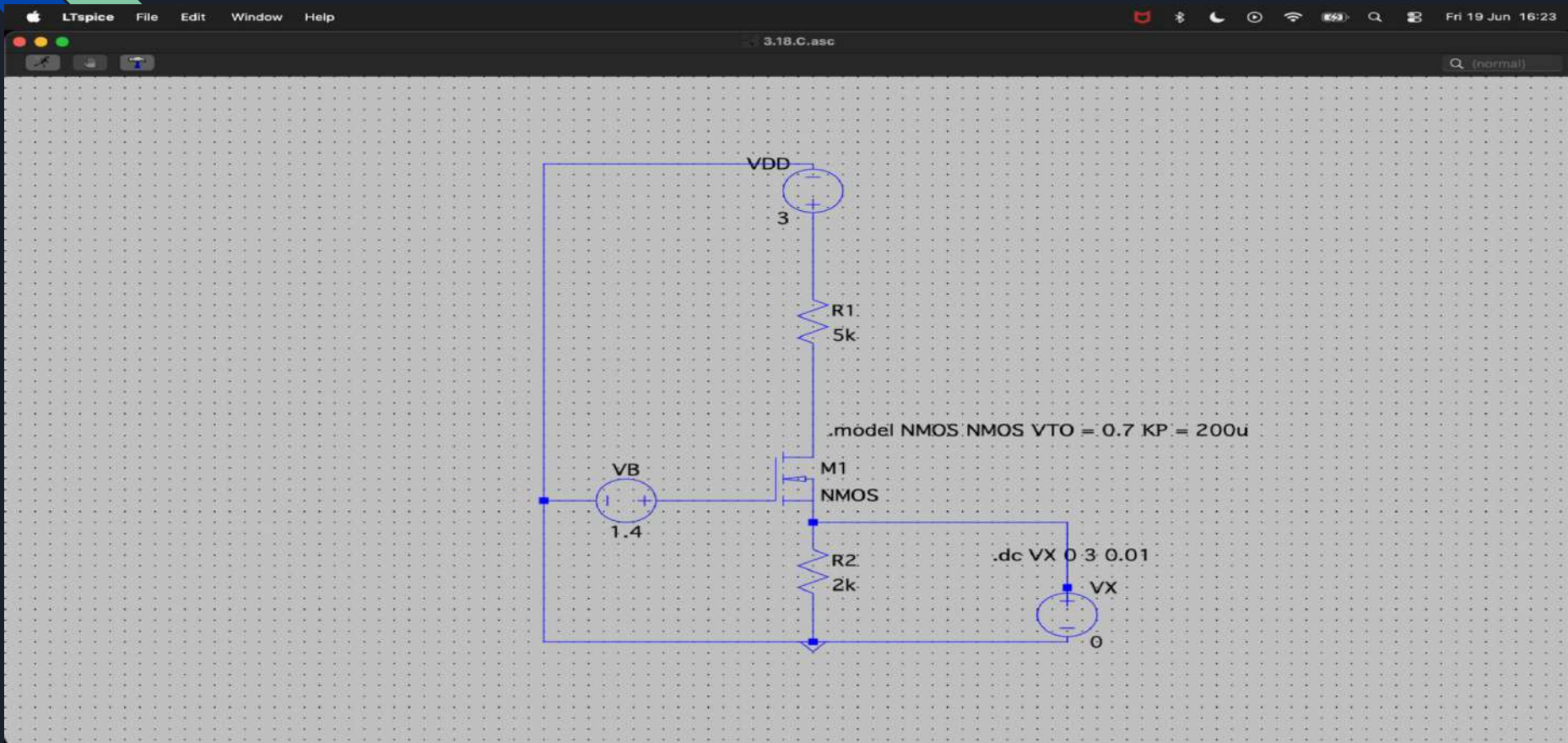
Also $I_D = 0$, when $I_D = I_D$

$\frac{V_X}{R_S} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_G - V_X - V_{th})^2$

We can find V_X from this equation



LTSPICE Schematic



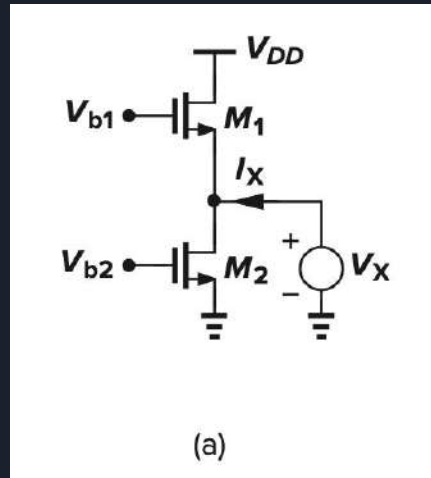
Simulation Results for I_x



Problem 3.19-a

3.19. Sketch I_X versus V_X for the circuits of Fig. 3.82 as V_X varies from 0 to V_{DD} . Identify important transition points.

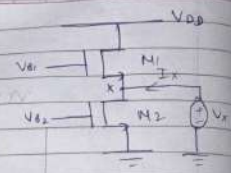
Diagram -



Handwritten Solution

3.19 YoV

Assuming $V_{D1} > V_{D2}$



Region	M1	M2
OFF	$V_X > V_{D1} - V_{th1}$	X
Triode	X	$V_X < V_{D2} - V_{th2}$
Saturation	✓	$V_X > V_{D2} - V_{th2}$

Initially when $V_X = 0$, $I_2 = 0$

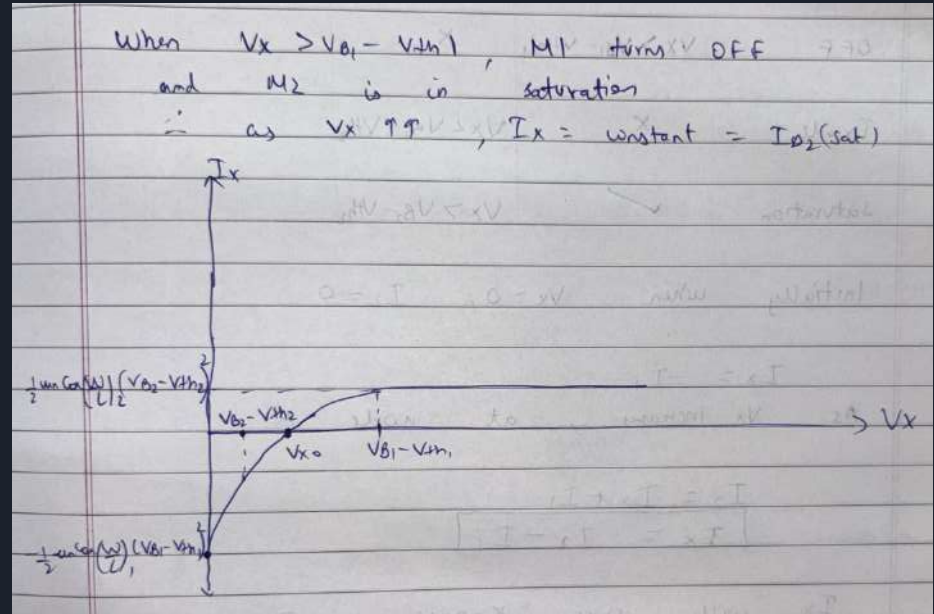
$I_X = -I_1$

As V_X increases, at node X,

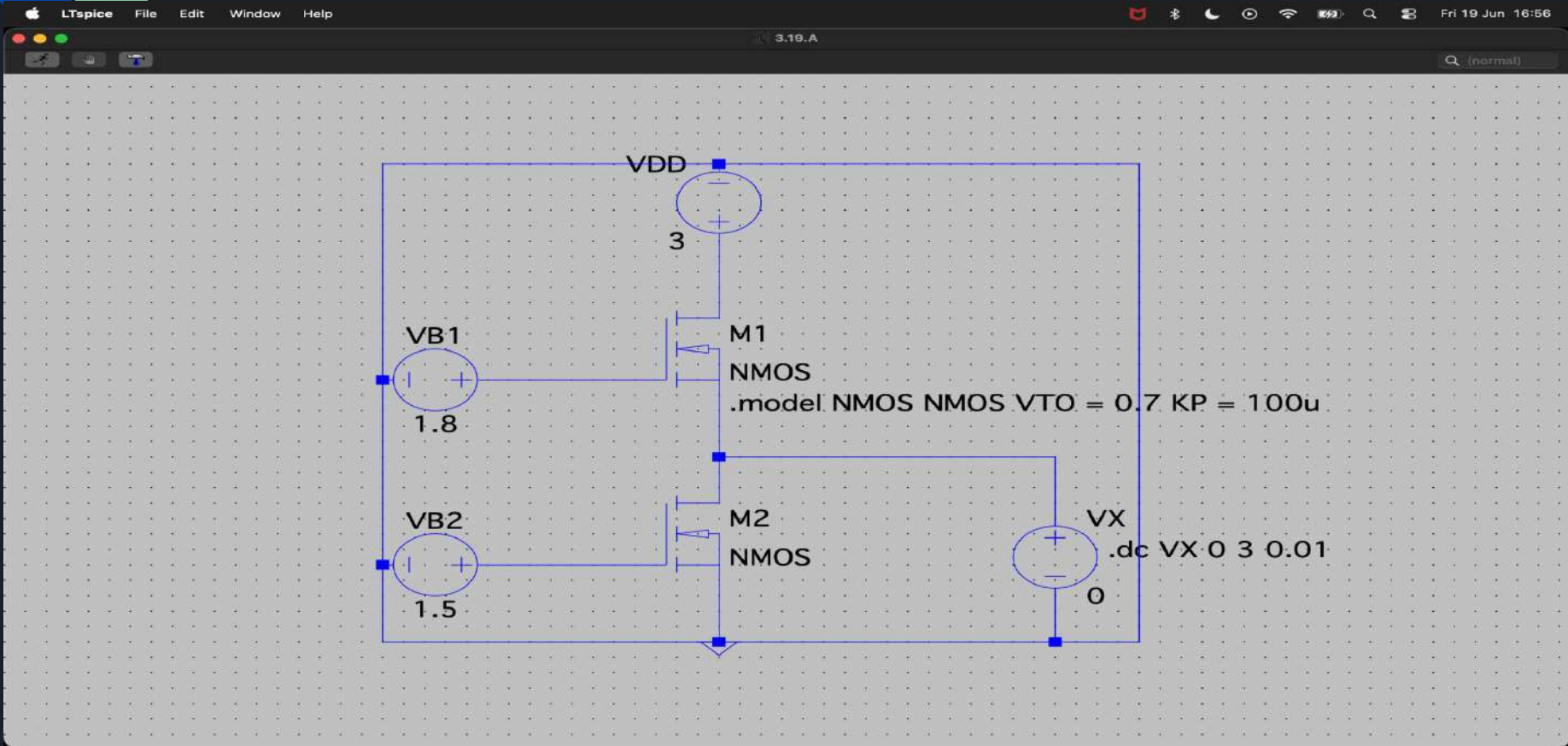
$$\begin{aligned} I_2 &= I_X + I_1 \\ I_X &= I_2 - I_1 \end{aligned}$$

I_X will cross X-axis, i.e. $I_X = 0$
when $I_1 = I_2$, this will occur when both are in saturation

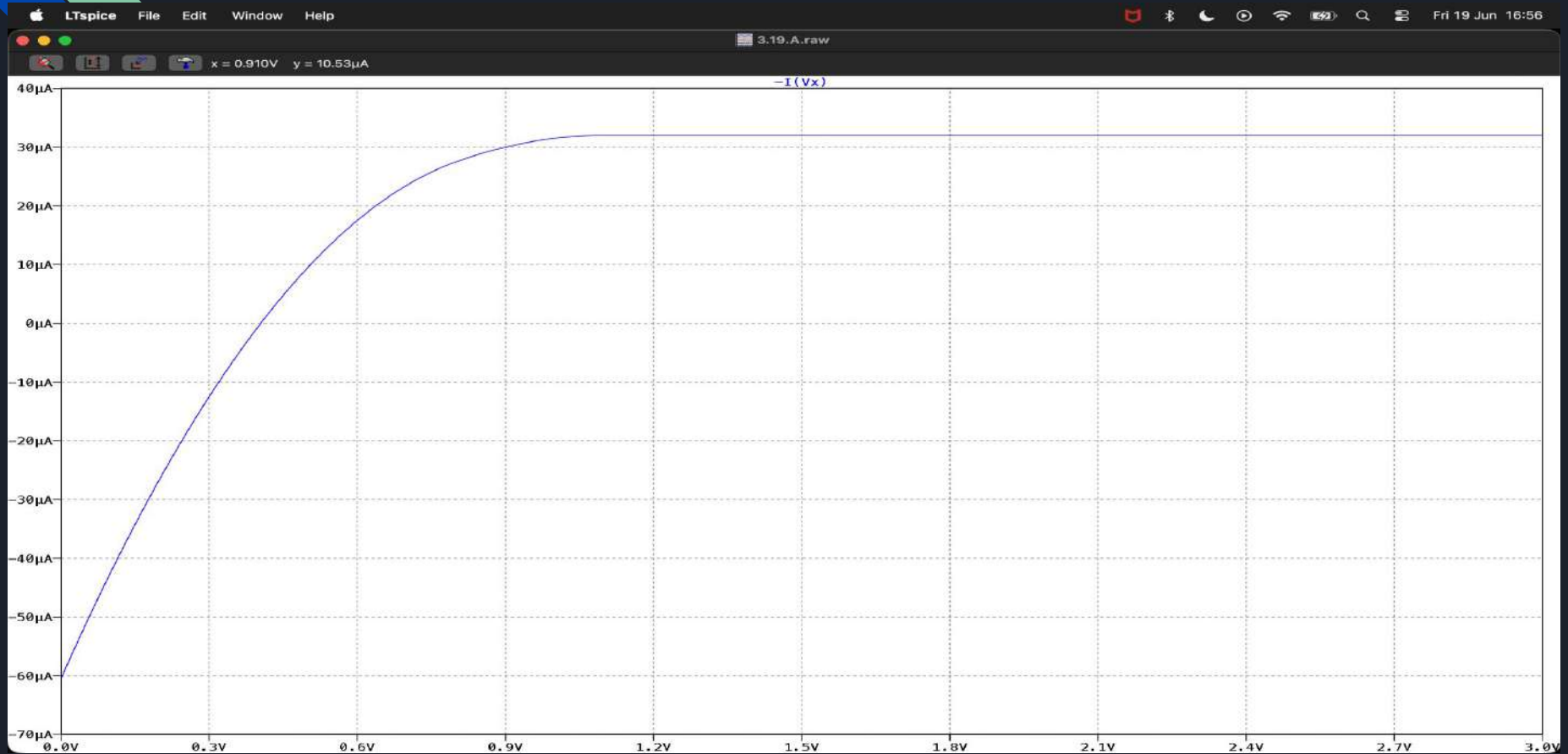
$$\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_{D1} - V_X - V_{th1})^2 = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_2 (V_{D2} - V_{th2})^2$$

$$V_{D1} - V_X - V_{th1} = \sqrt{\frac{(W/L)_2}{(W/L)_1}} (V_{D2} - V_{th2})$$


LTSPICE Schematic



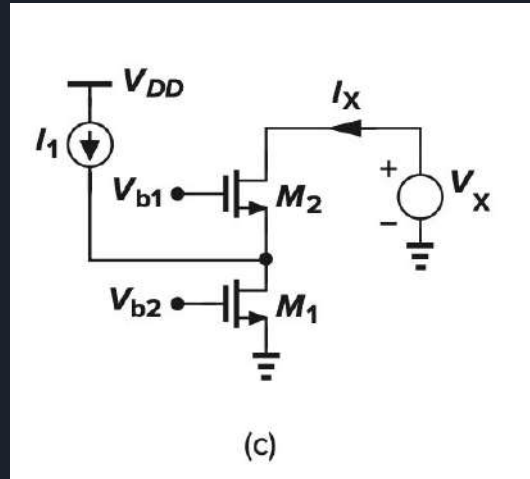
Simulation Results for I_x



Problem 3.19-c

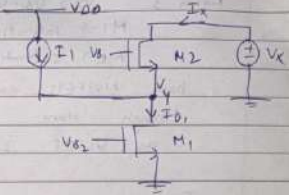
3.19. Sketch I_X versus V_X for the circuits of Fig. 3.82 as V_X varies from 0 to V_{DD} . Identify important transition points.

Diagram -



Handwritten Solution

2.19.4c4



When $V_x < V_y$
 Drain and source of M2 are interchanged
 $\therefore$ current flows in reverse direction till $V_x = V_y$

At node Y
 $I_1 + I_x = I_{D1}$
 $I_x = I_{D1} - I_1$

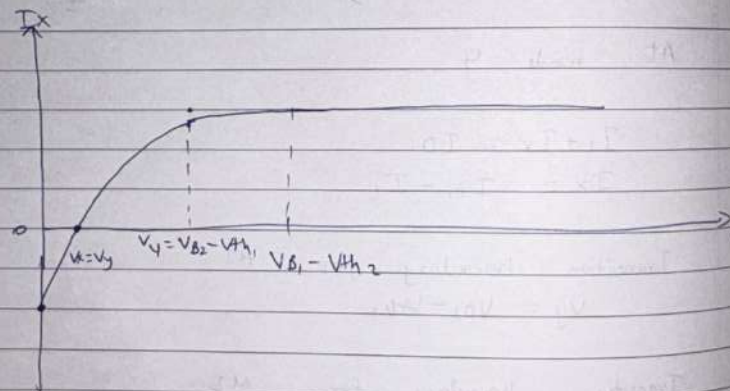
Transition boundary for M1
 $V_y = V_{D2} - V_{th1}$

Transition boundary for M2
 $V_x = V_{D1} - V_{th2}$

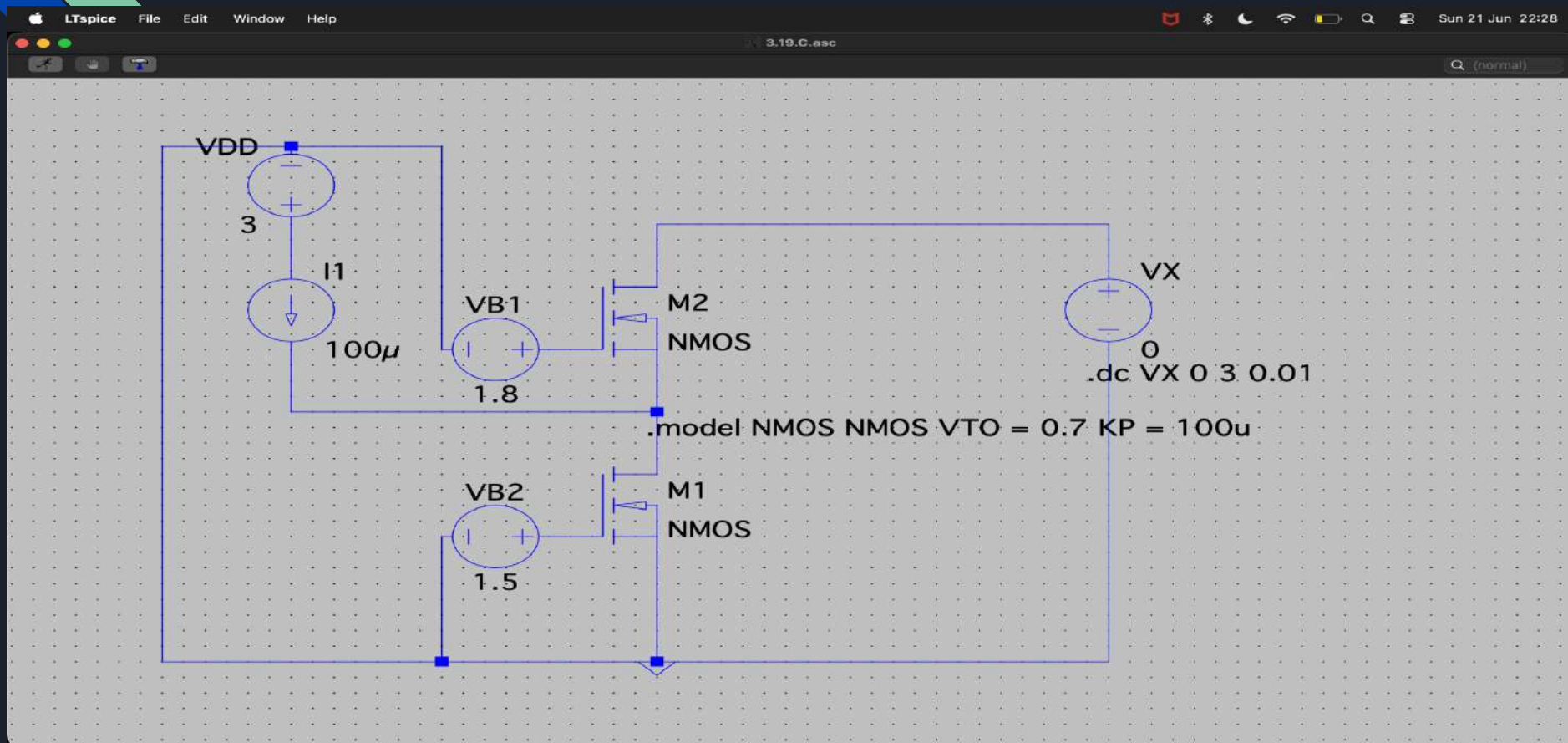
Assuming $V_{D1} > V_{D2}$
 Region 1, $V_y < V_{D2} - V_{th1}$
 M1 $\rightarrow$ triode
 M2 $\rightarrow$ triode
 $\therefore$ as $V_x \uparrow$, V_{D1} and V_{D2} increase
 since both are in triode region
 I_x also increases

Region 2, $V_y > V_{D2} - V_{th1}$, $V_x < V_{D1} - V_{th2}$
 M1 $\rightarrow$ saturation
 M2 $\rightarrow$ triode
 As $V_x \uparrow$, current I_x through M2 until it reaches the saturation current set by M1.

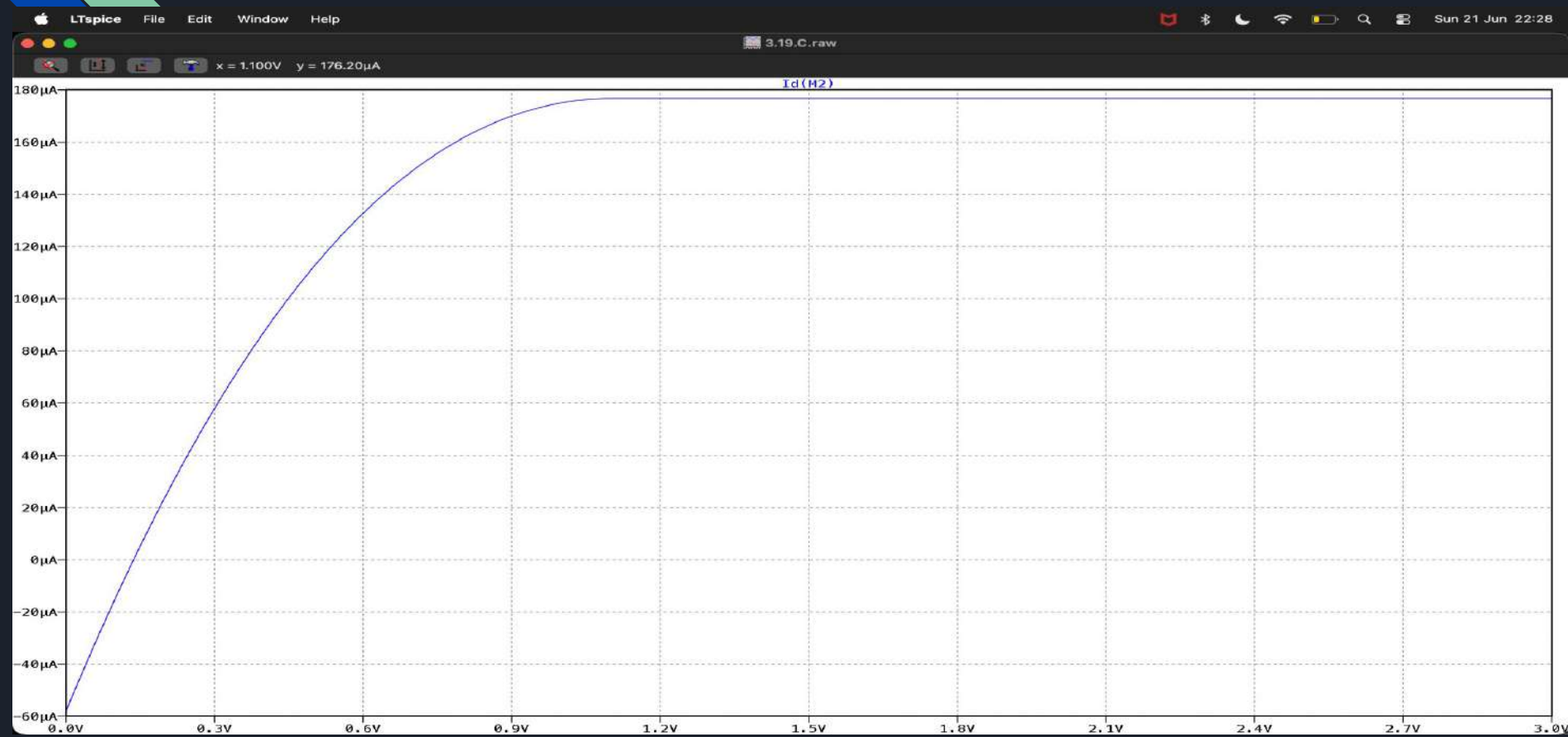
Region 3, $V_x > V_{D1} - V_{th2}$
 M1 $\rightarrow$ saturation
 M2 $\rightarrow$ saturation
 $\therefore$ both MOSFETs are in saturation current through them is constant
 $\therefore$ as $V_x \uparrow$, I_x remains constant



LTSPICE Schematic



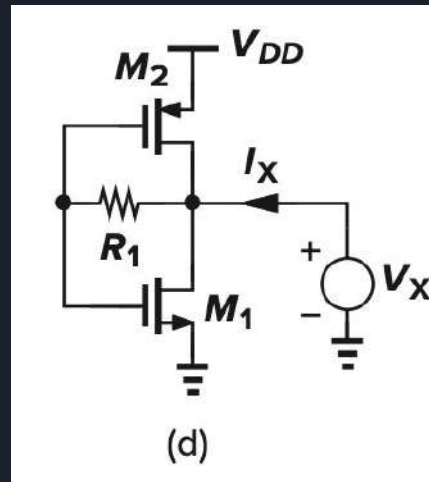
Simulation Results for Ix



Problem 3.19-d

3.19. Sketch I_X versus V_X for the circuits of Fig. 3.82 as V_X varies from 0 to V_{DD} . Identify important transition points.

Diagram -

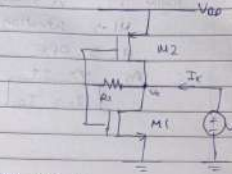


Handwritten Solution

314x14

$I_g = 0$ for MOSFET

$V_{g1} = V_{g2} = V_x$
Also $V_{d1} = V_{d2} = V_x$



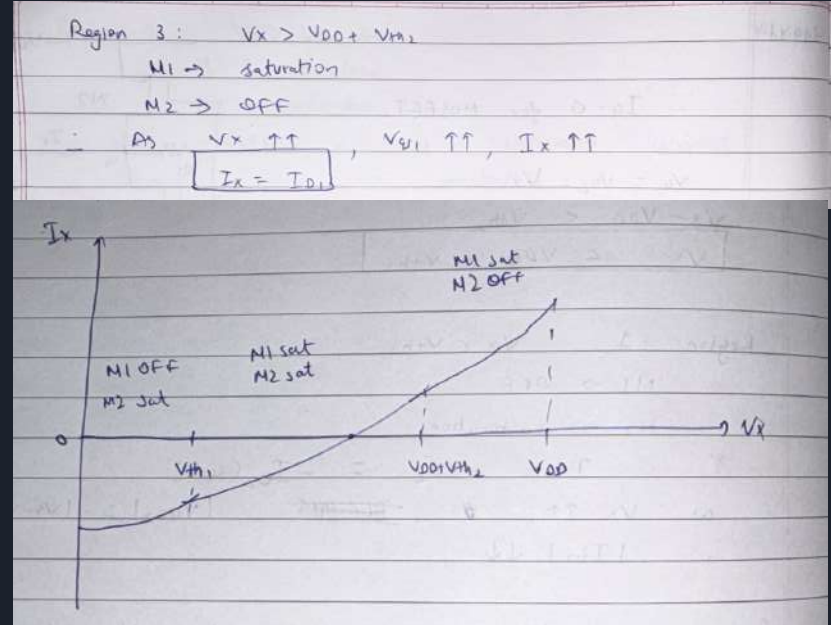
IF MOSFETs are ON, they will be in saturation

For M1 to turn ON
 $V_{g1} > V_{th1}$
 $V_x > V_{th1}$

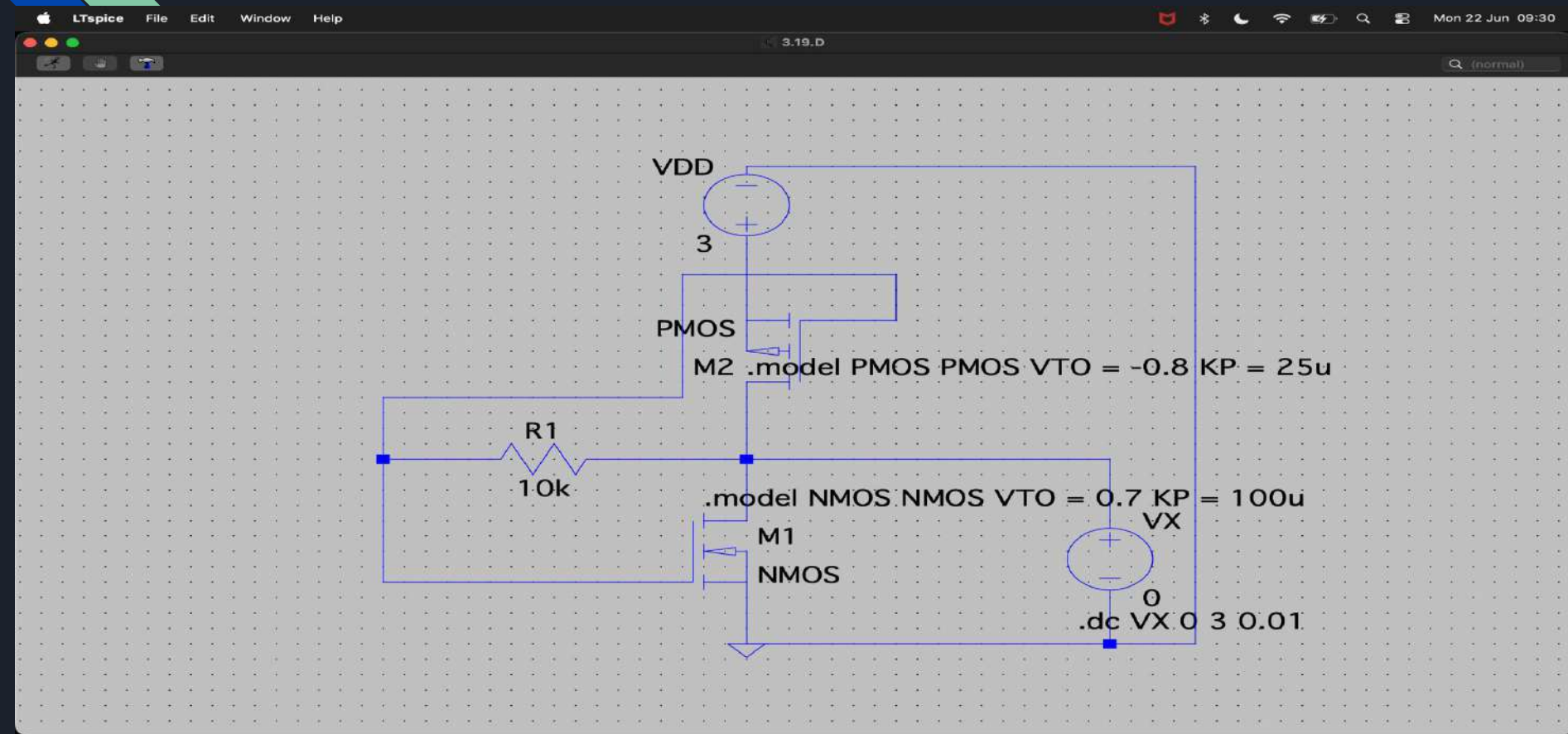
For M2 to turn ON
 $V_{g2} < V_{th2}$
 $V_x - V_{DD} < V_{th2}$
 $V_x < V_{DD} + V_{th2}$

Region 1: $V_x < V_{th1}$
M1 $\rightarrow$ OFF
M2 $\rightarrow$ saturation
 $I_x = -I_{D2} = -I_{D2(sat)}$
As $V_x \uparrow$, $|I_{D2}| \downarrow$

Region 2: $V_x > V_{th1}$ and $V_x < V_{DD} + V_{th2}$
M1 $\rightarrow$ saturation
M2 $\rightarrow$ saturation
As V_x increases, $I_{D1} \uparrow$, $|I_{D2}| \downarrow$,
but since M1 is stronger (wider channel)
 $I_x = (I_{D1} - I_{D2}) \uparrow$
Also in this region when $I_{D1} = I_{D2}$, $I_x = 0$
Operating channel length modulation:
 $\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_x - V_{th1})^2 = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_2 (V_{DD} - V_x - V_{th2})^2$



LTSPICE Schematic



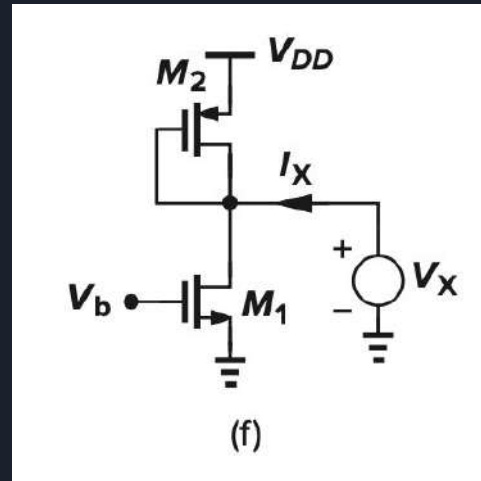
Simulation Results for I_x



Problem 3.19-f

3.19. Sketch I_X versus V_X for the circuits of Fig. 3.82 as V_X varies from 0 to V_{DD} . Identify important transition points.

Diagram -



Handwritten Solution

2.497f)

M2 will be in saturation whenever it is on

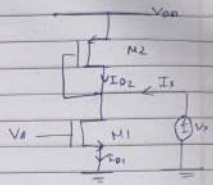
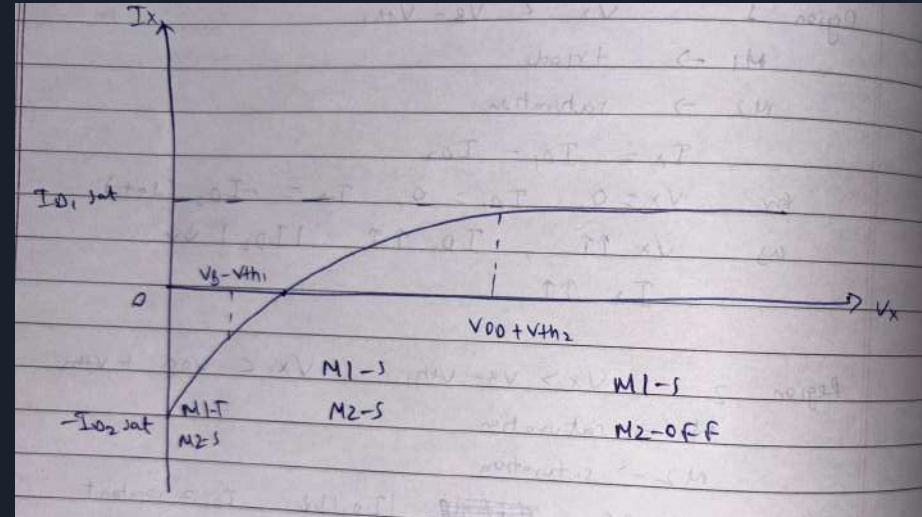
M2 will be: $V_{S2} > V_{th2}$
 $V_X > V_{DD} + V_{th2} \rightarrow M2 \text{ OFF}$

M1: triode: $V_X = V_S - V_{th1}$

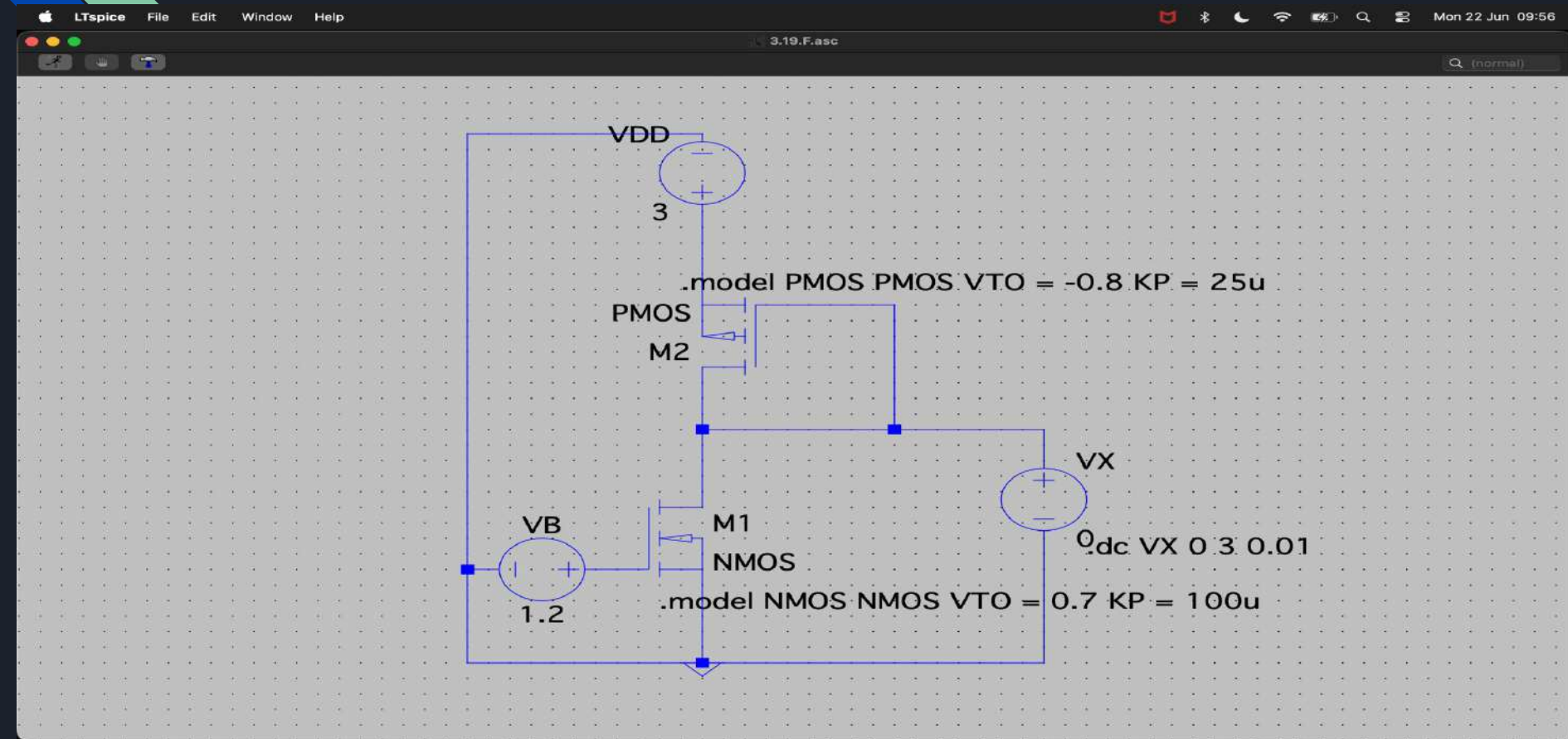
Region 1: $V_X < V_S - V_{th1}$
 M1 $\rightarrow$ triode
 M2 $\rightarrow$ saturation
 $I_A = I_{D1} - I_{D2}$
 for $V_X = 0$, $I_{D1} > 0$, $I_A = -I_{D2}$ (sat)
 as $V_X \uparrow$, $I_{D1} \uparrow$, $|I_{D2}| \downarrow$
 $\therefore I_A \uparrow$

Region 2: $V_X > V_S - V_{th1}$, $V_X < V_{DD} + V_{th2}$
 M1 $\rightarrow$ saturation
 M2 $\rightarrow$ saturation
 As $V_X \uparrow$, $|I_{D2}| \downarrow$, $I_{D1} = \text{constant}$
 $I_A = I_{D1} - I_{D2}$
 $I_A \uparrow$
 Also when $I_{D1} = I_{D2}$, $I_A = 0$
 $\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_S - V_{th1})^2 = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_2 (V_{DD} - V_X - V_{th2})^2$

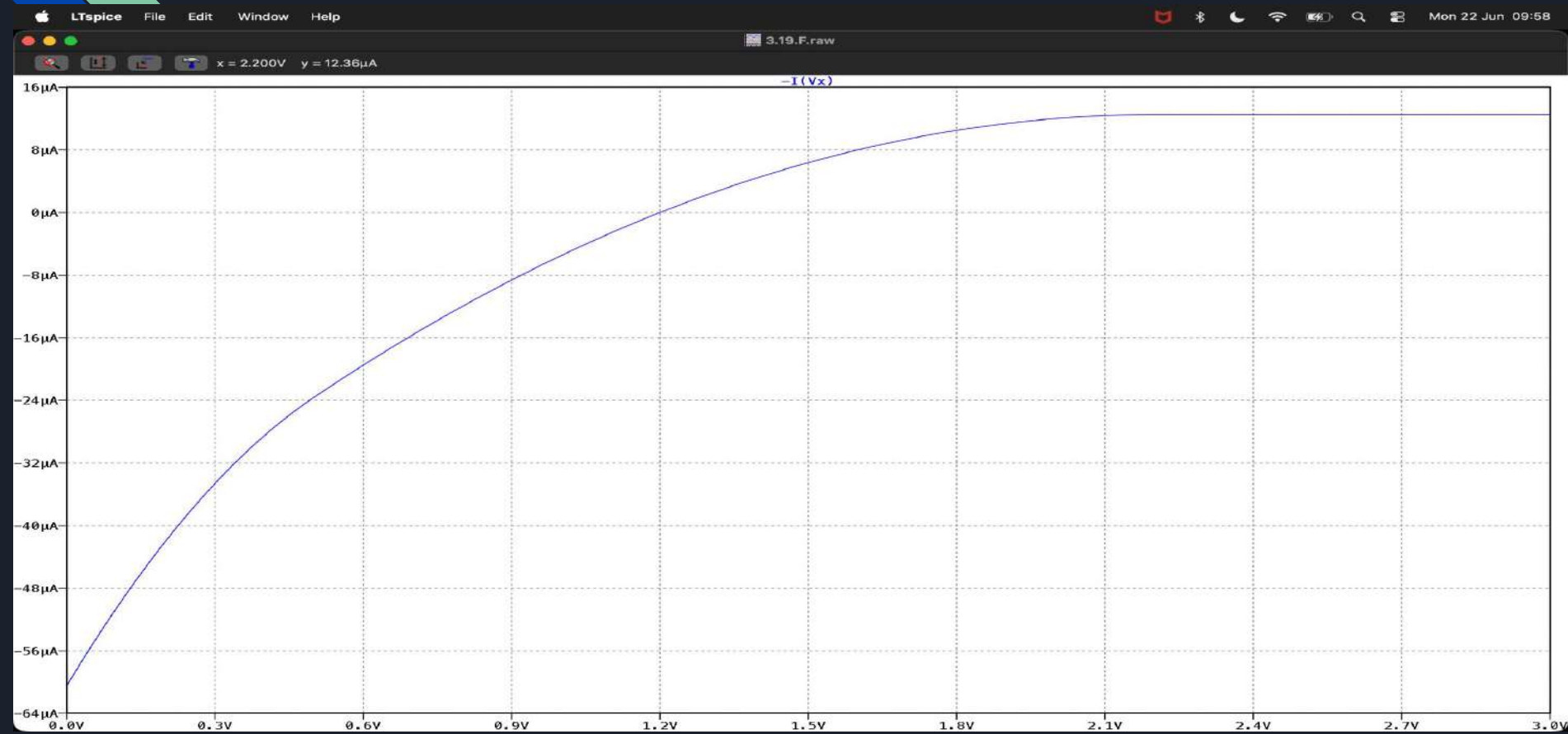
Region 3: $V_X > V_{DD} + V_{th2}$
 M1 $\rightarrow$ saturation
 M2 $\rightarrow$ off
 channel length modulation as $V_X \uparrow$, $I_{D1} = \text{constant}$
 $\therefore I_A = I_{D1} = \text{constant}$

LTSPICE Schematic



Simulation Results for I_x



THANK YOU

